

Eigenvalues of tetrahedral sums of Hermitian matrices

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Abstract

TODO: write an abstract

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1 Introduction

There are a number of operations involving the transformation of $d \times d$ Hermitian matrices under which it is possible to keep track of the set of eigenvalues, $\lambda(X)$, of a given Hermitian matrix X . For example, eigenvalues of Hermitian matrices are invariant under conjugation by unitary matrices, $\lambda(X) = \lambda(UXU^\dagger)$, covariant under taking positive powers $\lambda(X^p) = \lambda(X)^p$, covariant under multiplication by real numbers, $\lambda(rX) = r\lambda(X)$, and covariant under the addition of scalar matrices, $\lambda(X + rI) = \lambda(X) + (r, \dots, r)$. The operation of *addition* of Hermitian matrices is however not an operation of this type. Specifically, the eigenvalues $\gamma = \lambda(A+B)$ of the sum of two matrices cannot be determined from the eigenvalues $\alpha = \lambda(A)$ and $\beta = \lambda(B)$ of the summands *alone*; additional knowledge about the eigenvectors involved is typically required. Nevertheless, there are numerous mathematical situations wherein knowledge about the eigenvectors involved is lacking, yet it remains desirable to determine the set of *possible* eigenvalues of a sum given the eigenvalues of the summands.

The problem of determining the subset of admissible eigenvalues has come to be known as *Horn's problem*; it is named after Horn who, in 1962, conjectured a solution to the problem in the form of an inductively defined and finite set of homogeneous linear inequalities, known as the *Horn inequalities* describing a polytope of admissible values for (α, β, γ) [Hor62]. In 1998, Klyachko provided a solution to Horn's problem [Kly98] which, when combined with a proof of the *saturation conjecture* due to Knutson and Tao [KT99], constitutes a proof of Horn's original conjecture [Ful98].

TODO: Summarize reviews [Ful00; Knu00; Ful00; Bha01].

TODO: Summarize multiplicative variations for products of unitaries [Bel08; BK16] and products of unimodular matrices and the proof of Thompson's conjecture [Kly00].

In this paper, we consider a generalization of the additive version of Horn's problem which we refer to as a *linear eigenvalue constraint problems* or *LECPs*. A linear eigenvalue constraint problem (LECP) may be viewed as a generalization of a homogeneous system of linear equations wherein real-valued variables are replaced with Hermitian-matrix-valued variables which are additionally subject to eigenvalue constraints.

Problem 1 (Linear eigenvalue constraint problem (LECP)). Let $\Gamma \in \mathbb{R}^{p \times q}$ be a given $p \times q$ real-valued matrix and let $\Lambda \in \mathbb{R}^{q \times d}$ be a given $q \times d$ real-valued matrix. If there exists a q -tuple $(X_1, \dots, X_q)$ of $d \times d$ Hermitian matrices satisfying (i) for all $i \in \{1, \dots, p\}$ the *linear constraint*

$$\sum_{j=1}^q \Gamma_{i,j} X_j = 0, \quad (1)$$

and (ii) for all $j \in \{1, \dots, q\}$ the *eigenvalue constraint*

$$\lambda(X_j) = (\Lambda_{j,1}, \dots, \Lambda_{j,d}), \quad (2)$$

then the matrix of eigenvalues Λ is said to be *admissible* for Γ . The *linear eigenvalue constraint problem* for Γ is to describe or characterize the set of all admissible spectra for Γ .

TODO: Describe how the $d = 1$ case reduces to ordinary homogeneous systems of linear equations, how one may always taken eigenvalues to be traceless if desired, how the $d = 2$ traceless case with $\{-1, 0, +1\}$ -valued coefficients for Γ reduces to three-dimensional Euclidean geometric constraint problems [SJS18] [AW98], how the $p = 1$ case reduces to the additive Horn problem and how some of the tree-like $p \geq 2$ cases also reduce to the ordinary Horn problem also.

Throughout this paper, a considerable focus will be placed on the following particular instance of a linear eigenvalue constraint problem involving six Hermitian matrices subject to four constraints which are derived from the four faces of a tetrahedron with six edges as depicted in Fig. 1.

prob:tetrahedral_problem

Problem 2 (The tetrahedral eigenvalue constraint problem). A sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in (\mathbb{R}_{\text{dec}}^d)^{\times 6}$ of candidate eigenvalues are said to be *tetrahedral* if there exists a corresponding sextuple (A, B, C, D, E, F) of $d \times d$ Hermitian matrices satisfying (i) the four linear constraints

{eq:tetrahedral_linear_constraints}

$$A + B = C, \quad B + D = F, \quad D + C = E, \quad A + F = E, \quad (3)$$

and (ii) the six eigenvalue constraints

{eq:tetrahedral_eigenvalue_constraints}

$$\lambda(A) = \alpha, \quad \lambda(B) = \beta, \quad \lambda(C) = \gamma, \quad \lambda(D) = \delta, \quad \lambda(E) = \epsilon, \quad \lambda(F) = \phi. \quad (4)$$

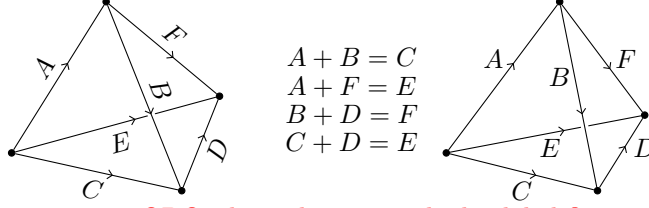


Figure 1: The six Hermitian operators appearing in the tetrahedral eigenvalue constraint problem (Problem 2), and denoted by (A, B, C, D, E, F) above, are subject to four linear constraints corresponding to the four faces of the tetrahedron with coefficients determined by the edge orientations depicted here. {eq:tetrahedron_labelling_scheme}

There are two reasons for our focus on the tetrahedral eigenvalue constraint problem. The first reason simply that it exhibits many of the challenges typical of more general linear eigenvalue constraint problems that are not already challenges present in the original additive Horn problem.

The second reason for our interest in the tetrahedral eigenvalue constraint problem stems from the fact that an elegant, complete solution already exists for the case of 2×2 Hermitian matrices. In order to describe this solution, first note that to solve a LECP it suffices to solve the specific case where the Hermitian operators, and thus eigenvalues, are assumed traceless (see Section 2.1). Moreover, in the context of a two-dimensional LECP, each traceless 2×2 matrix, X , may be identified with a three-dimensional vector $(x_1, x_2, x_3) \in \mathbb{R}^3$ using the standard Pauli-matrix basis:

$$X = x_1\sigma_1 + x_2\sigma_2 + x_3\sigma_3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}. \quad (5)$$

The eigenvalues of X are then determined by the Euclidean length of the vector (x_1, x_2, x_3) :

$$\lambda(X) = \lambda \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix} = (\|x\|, -\|x\|), \quad \text{where,} \quad \|x\| = \sqrt{x_1^2 + x_2^2 + x_3^2}. \quad (6)$$

Now, for the special case of the tetrahedral eigenvalue constraint problem, one can assume the sextuple of eigenvalues are determined by a sextuple of non-negative real numbers $(a, b, c, d, e, f) \in \mathbb{R}_{\geq 0}^6$ such that {eq:traceless_two_dim_tetra}

$$\alpha = (a, -a), \quad \beta = (b, -b), \quad \gamma = (c, -c), \quad \delta = (d, -d), \quad \epsilon = (e, -e), \quad \phi = (f, -f). \quad (7)$$

Using this parameterization, it can be shown that the constraints imposed in Problem 2 precisely describe the space of all possible edge-lengths (a, b, c, d, e, f) of a genuine Euclidean tetrahedron in $\mathbb{R}^3$ [Ric02]; namely, the conjunction of a quadruplet of triangle-inequalities, {eq:quadruplet_triangle_constraints}

$$|a - b| \leq c \leq a + b, \quad |b - d| \leq f \leq b + d, \quad |d - c| \leq e \leq d + c, \quad |a - f| \leq e \leq a + f, \quad (8)$$

together with a single non-linear constraint of the form

$$D(a, b, c, d, e, f) = \frac{1}{288} \det \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & a^2 & e^2 & c^2 \\ 1 & a^2 & 0 & f^2 & b^2 \\ 1 & e^2 & f^2 & 0 & d^2 \\ 1 & c^2 & b^2 & d^2 & 0 \end{pmatrix} \geq 0. \quad (9) \quad \text{span style="color: blue;">{eq:volume_positive_constraint}}$$

The quantity $D(a, b, c, d, e, f)$ is known as a Cayley-Menger determinant and, when positive, is equal to square of the volume of the corresponding tetrahedron [SJS18]. Figure 2 depicts a three-dimensional projection of the solution space described by Eq. (8) and Eq. (9).

Lemma 1 (Tetrahedral shifting). *Let $(\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}, \tilde{\delta}, \tilde{\epsilon}, \tilde{\phi})$ be a given sextuple of candidate eigenvalues and let $x, y, z \in \mathbb{R}$ be given scalars and define shifted candidate eigenvalues by the formula*

$$\begin{aligned} \alpha &:= \tilde{\alpha} + x, & \beta &:= \tilde{\beta} + y, & \gamma &:= \tilde{\gamma} + x + y, \\ \delta &:= \tilde{\delta} + z, & \epsilon &:= \tilde{\epsilon} + x + y + z, & \phi &:= \tilde{\phi} + y + z. \end{aligned} \quad (10) \quad \text{span style="color: blue;">{eq:shifted_tetrahedral_eigenvalues}}$$

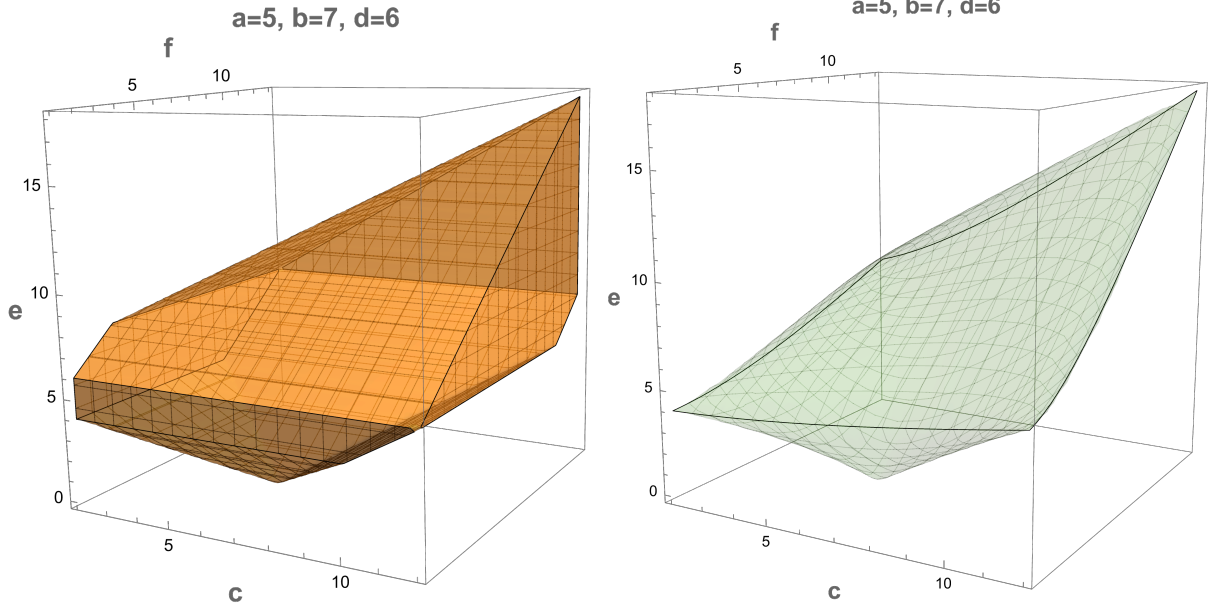


Figure 2: Projections of admissible side-lengths (a, b, c, d, e, f) for Euclidean tetrahedron along the coordinates $(a, b, d) = (5, 7, 6)$ (Left) The polyhedron of values (c, e, f) satisfying the triangle inequalities for each face Eq. (8). (Right) The set of all values (c, e, f) satisfying both Eq. (8) and Eq. (9). [tetrahedron_moduli_space_example](#)

Then the shifted eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral if and only if the unshifted eigenvalues are tetrahedral. Moreover if the unshifted eigenvalues are tetrahedral and (x, y, z) are sufficiently large, i.e.

$$x \geq \min(\tilde{\alpha}), \quad y \geq \min(\tilde{\beta}), \quad z \geq \min(\tilde{\delta}), \quad (11)$$

then the shifted spectra are both tetrahedral and non-negative.

In this paper we prove the following theorem which serves as a necessary (albeit insufficient) condition for a given collection of eigenvalues to be tetrahedral which holds for arbitrary d . [thm:SSA](#)

Theorem 2. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be a tetrahedral sextuple of non-negative eigenvalues. Then the following inequality necessarily holds: [{eq:SSA_like_ineq}](#)

$$s(\gamma) + s(\phi) - s(\beta) - s(\epsilon) \geq 0, \quad (12)$$

where $s(\lambda)$ is defined for non-negative $\lambda = (\lambda_1 \geq \dots \geq \lambda_d \geq 0)$ by the formula

$$s(\lambda) := - \sum_{i=1}^{\ell(\lambda)} \lambda_i \log \lambda_i. \quad (13) \quad \text{[{eq:unnorm_shannon_intro}](#)}$$

Remark 1. For 2×2 Hermitian matrices it is possible to visualize the set of eigenvalues (in the same (a, b, c, d, e, f) coordinates used above) which are known to be non-tetrahedral on the basis that they violate Eq. (12). Specifically, consider the *minimal* shifting of the traceless eigenvalues in Eq. (7), i.e.,

$$\begin{aligned} \alpha &= (2a, 0), & \beta &= (2b, 0), & \gamma &= (a + b + c, a + b - c), \\ \delta &= (2d, 0), & \epsilon &= (a + b + d + e, a + b + d - e), & \phi &= (b + d + f, b + d - f). \end{aligned} \quad (14)$$

With respect to these coordinates, Eq. (12) becomes

$$\begin{aligned} & - (a + b + c) \log(a + b + c) - (a + b - c) \log(a + b - c) \\ & - (b + d + f) \log(b + d + f) - (b + d - f) \log(b + d - f) \\ & + (a + b + d + e) \log(a + b + d + e) + (a + b + d - e) \log(a + b + d - e) \\ & + (2b) \log(2b) \geq 0 \end{aligned} \quad (15) \quad \text{[{eq:SSA_like_d_two}](#)}$$

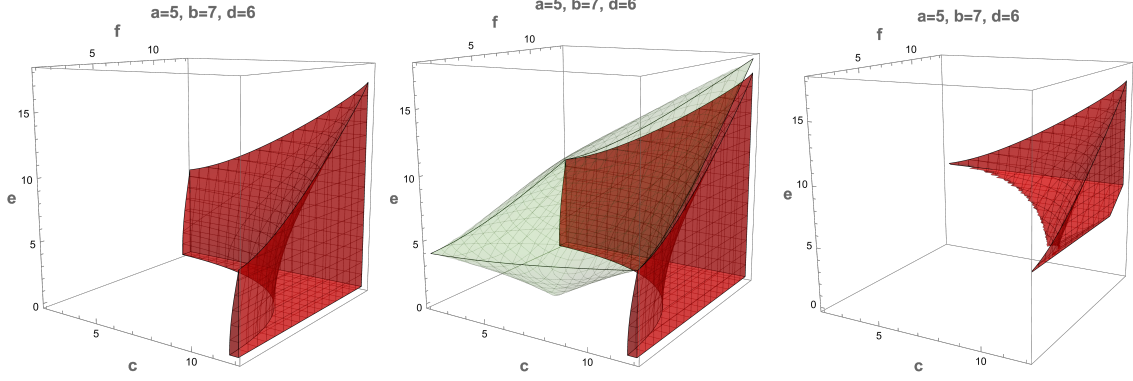


Figure 3: (Left) The region for which (c, e, f) violates Eq. (15) with $(a, b, c) = (5, 7, 6)$. (Middle) Comparing the region of tetrahedral (c, e, f) with those which violate Eq. (15). (Right) The values of (c, e, f) satisfying Eq. (8) but violating Eq. (15). fig:ssa_extra_mileage

See Fig. 3 for a depiction of the region violating Eq. (15).

Remark 2. At this stage, it is worth comparing the inequality in Eq. (12) to a similar inequality derived by Christandl, Şahinoğlu, and Walter [CSW18], which holds for tetrahedral eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ which are non-negative and sufficiently small in the sense that $\lambda_1 + |\lambda| \leq 1$:

$$h(|\gamma|) + |\gamma| S(\bar{\gamma}) + h(|\phi|) + |\phi| S(\bar{\phi}) - h(|\beta|) - |\beta| S(\bar{\beta}) - h(|\epsilon|) - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad \text{\textcolor{blue}{eq:ssa_like_CSW}} \quad (16)$$

The inequality appearing in Eq. (12) is equivalent to the following by Eq. (156):

$$-|\gamma| \log |\gamma| + |\gamma| S(\bar{\gamma}) - |\phi| \log |\phi| + |\phi| S(\bar{\phi}) + |\beta| \log |\beta| - |\beta| S(\bar{\beta}) + |\epsilon| \log |\epsilon| - |\epsilon| S(\bar{\epsilon}) \geq 0. \quad \text{\textcolor{blue}{eq:ssa_like_ineq_equiv}} \quad (17)$$

The difference between Eq. (16) and Eq. (17) is captured by a formal substitution replacing the binary entropy $h(|\lambda|)$ with the expression $-|\lambda| \log |\lambda|$. We note that it is possible to derive Eq. (17) from Eq. (16) by a downscaling argument. Specifically, while the set of tetrahedral eigenvalues is closed under scaling all eigenvalues by a non-negative r ,

$$(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \mapsto (r\alpha, r\beta, r\gamma, r\delta, r\epsilon, r\phi), \quad \text{\textcolor{blue}{eq:tetrahedral_scaling}} \quad (18)$$

the form and positivity of Eq. (16) is *not* invariant under scaling by r . Define q_r by the formula

$$q_r := r^{-1} (h(|r\gamma|) + h(|r\phi|) - h(|r\beta|) - h(|r\epsilon|)). \quad (19)$$

Letting $r \in (0, 1)$ be positive and small, we see that

$$r^{-1} h(|r\lambda|) = |\lambda| (1 - \log r) - |\lambda| \log |\lambda| + O(r^2) \quad (20)$$

which implies the following limit holds when $|\gamma| + |\phi| = |\beta| + |\epsilon|$:

$$q_0 := \lim_{r \rightarrow 0} q_r = -|\gamma| \log |\gamma| - |\phi| \log |\phi| + |\beta| \log |\beta| + |\epsilon| \log |\epsilon|. \quad (21)$$

In summary, by scaling down the eigenvalues in Eq. (16) by $r \in (0, 1)$ according to Eq. (18) and then scaling up Eq. (16) by multiplication by r^{-1} and taking the limit as $r \rightarrow \infty$, one recovers Eq. (17).

TODO: Based upon numerics, TC conjectures that Eq. (12) is always strictly stronger than Eq. (16). Further analysis in combination with the trace equalities reveals

$$x \in [0, 1] \implies -(1-x) \log(1-x) - (x-x^2) \in [0, 0.161903\dots], \quad (22)$$

which in turn implies

$$q_r - q_0 = r |\alpha| |\delta| + O(r^2) \quad (23)$$

which implies $q_r > q_0$ and least for sufficiently small r . With the knowledge that $(x, y, x+y, z, x+y+z, y+z)$ are tetrahedral, Theorem 2 for $d = 1$ implies the following non-trivial inequality for $x, y, z \geq 0$:

$$-(x+y) \log(x+y) - (y+z) \log(y+z) + (x+y+z) \log(x+y+z) + (y) \log(y) \geq 0. \quad (24)$$

I strongly suspect that the difference $r(q_r - q_0)$ can be written in the above form using something like $x = 1 - r |\alpha|$, $y = 1 - r |\beta|$ and $z = 1 - r |\delta|$ thus proving that $q_r \geq q_0$ always. More work is needed!

1.1 Notational conventions

TODO: Review and refresh the notational conventions based on the final draft! Throughout this paper, $d \in \mathbb{N}_{>0}$ will denote the size of the matrices under consideration. We also exclusively work with matrices with coefficients in the field of complex numbers $\mathbb{C}$, so for notational brevity we write $\text{GL}(d)$ as shorthand for $\text{GL}(d, \mathbb{C})$. Weights for $\text{GL}(d)$ will be denoted using Greek letters and will be identified with d -tuples of integers, e.g. $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{Z}^d$. The weight lattice for $\text{GL}(d)$ is denoted by $\mathbb{Z}^d \simeq \mathbb{Z}^d$ in the usual way; a weight $\alpha \in \mathbb{Z}^d$ is dominant if $\alpha_1 \geq \alpha_2 \geq \dots \geq \alpha_d$ and the set of all dominant weights for $\text{GL}(d)$ is denoted by $\mathbb{Z}_{\text{dec}}^d$. Irreducible highest weight representations of $\text{GL}(d)$ with weight $\lambda \in \mathbb{Z}^d$ are denoted by V_λ and irreducible representations of S_n of highest weight $\lambda \in \Upsilon_{n,\infty}$ are denoted by W_λ .

The set of all $d \times d$ complex Hermitian matrices will be denoted by $\mathcal{H}(d)$ while the set of all possible eigenvalues of a $d \times d$ complex Hermitian matrix is denoted by $\mathbb{R}_{\text{dec}}^d \subset \mathbb{R}^d$. The set of $d \times d$ Hermitian matrices X with eigenvalues $\text{eig}(X) = (x_1, \dots, x_d) \in \mathbb{R}_{\text{dec}}^d$ will be denoted by $\mathcal{O}_x$.

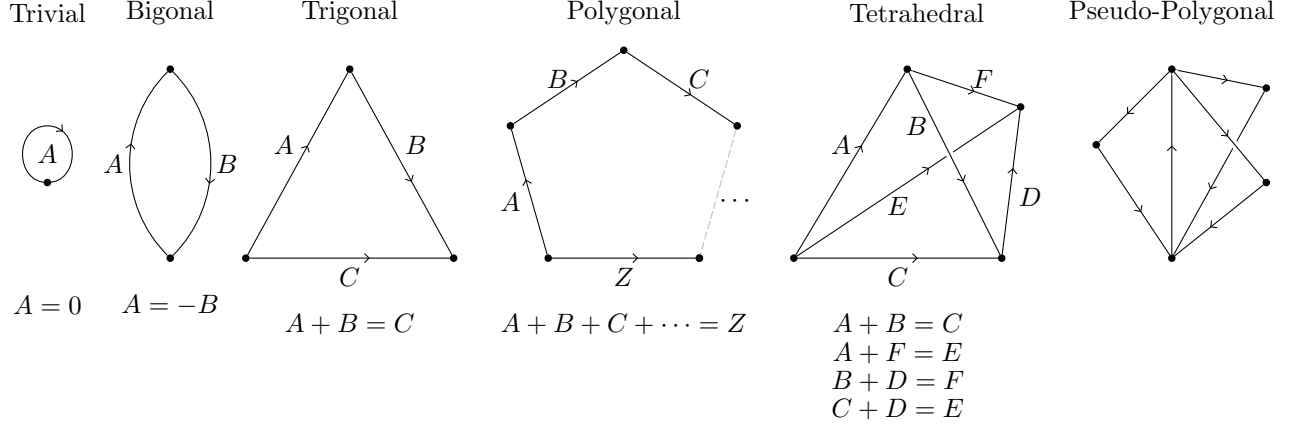


Figure 4: Examples of graphical Horn problems.

2 Generalizations of Horn's problem

This section will include a formal statement of the problem under investigation, its relation to Horn's original problem and other generalizations of Horn's problem. This section will hopefully briefly describe what is known about the eigenvalues of triangular sums of matrices. Regarding tetrahedral spectra, we will describe many of the symmetries of the problem statement and various “gauge” choices that one could make. Additional generalizations of the problem statement, like to more matrices, linear combinations, etc. will be mentioned here. It will be important to mention the manner in which the analogue of the “Thompson-Klyachko” theorem does not hold, i.e. singular values of tetrahedral products are not the exponentials of eigenvalues of tetrahedral sums in general.

2.1 Graphical Horn problems

`sec:linear_eigenvalue_constraint_problems`

Definition 1. Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph with vertex set $\mathcal{V}$ and edge set $\mathcal{E}$ together with source and target maps $s, t : \mathcal{E} \rightarrow \mathcal{V}$. Let d be a positive integer. A *spectral map* (for $\mathcal{G}$ of dimension d) is a function $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ assigning to each edge $e \in \mathcal{E}$ in the directed graph a d -tuple of real numbers $\lambda(e) = (\lambda_1(e), \dots, \lambda_d(e))$. The map $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is said to be *realizable (by Hermitian matrices)* if there exists a map $H : \mathcal{E} \rightarrow \mathcal{H}(d)$ assigning to each edge $e \in \mathcal{E}$ a $d \times d$ Hermitian matrix $H(e)$ such that

- for each edge $e \in \mathcal{E}$, $H(e)$ has eigenvalues $\lambda(e)$,

$$\text{eig}(H(e)) = \lambda(e), \quad \text{\texttt{\{eq:spec_cond\}}} \quad (25)$$

- for each cycle $\ell \subseteq \mathcal{E}$ with orientation map $\gamma_\ell : \ell \rightarrow \{\pm 1\}$,

$$\sum_{e \in \ell} \gamma_\ell(e) H(e) = 0. \quad \text{\texttt{\{eq:cycle_cond\}}} \quad (26)$$

Alternatively, one can equivalently encode the constraints imposed by Eq. (26) in terms of a $\{-1, 0, +1\}$ -valued $L \times |E|$ matrix (where L is the number of (generating) cycles in $\mathcal{G}$), denoted by $\Gamma(\mathcal{G})$, where

$$\Gamma(\mathcal{G})_{\ell, e} = \begin{cases} 0 & e \notin \ell, \\ \gamma_\ell(e) & e \in \ell. \end{cases} \quad (27)$$

In terms of $\Gamma(\mathcal{G})$, Eq. (26) reads for all cycles ℓ :

$$\sum_{e \in \mathcal{E}} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0 \quad (28)$$

Problem 3 (Graphical Horn problem). Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a finite directed graph, let d be a positive integer. The *graphical Horn problem* is to characterize the set of realizable spectral maps $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ for $\mathcal{G}$ of dimension d .

There might be an important connection to the their of quivers and representations of quivers. In particular, the notion of a morphism between a representation of a quiver in terms of operators on vertices forming computing squares models precisely the kinds of continuous symmetries which may be present in a given graphical Horn problem. Indeed, the spectral data were are ultimately interested in is equivalent to the orbit under such “local” isomorphisms (i.e. conjugation by isomorphisms).

TC: It turns out that the only relevant data for statement of a graphical Horn problem is the matrix $\Gamma(\mathcal{G})$; perhaps its best to drop reference to $\mathcal{G}$ altogether and just have the problem statement depend on a matrix Γ ? Moreover, one could in principle take the matrix to be $\mathbb{R}$ -valued instead of $\{-1, 0, +1\}$ -valued, although that would make the connection to the representation theory even weaker. Generalizing to $\mathbb{Z}$ -valued Γ is also essentially the same I think... One downside of dropping the graph $\mathcal{G}$ is that the matrix Γ loses the non-abelian information about the ordering of the elements in a cycle; this is problematic when discussing the “quantum” generalizations to multiplicative Horn-type problems. One upside of dropping the graph $\mathcal{G}$, is that it becomes possible to ignore certain cycles of the graph (by dropping rows of Γ) and/or additionally including constraints non-easily associated to cycles of a common graph (like $A + C = Q, B + Q = L$ constraints for the tetrahedral problem).

Another option is to consider dual graphs. The proposal would be to attribute a cycle constraints to oriented vertices with incoming and outgoing vertices. This would make the connection to the networks built in the category of representations of $\text{GL}(d)$ much more apparent. TC: I missed this presentation because I fixated on the self-dual tetrahedron.

In the case of traceless 2×2 Hermitian matrices, the corresponding graphical Horn problem becomes equivalent to the problem of the existence of an embedding of an edge-weighted graph into $\mathbb{R}^3$. See [AW98] for a problem statement for embeddings into $\mathbb{R}^r$ and a constructive algorithm for producing a particular embedding when one exists.

There are a few operations which preserve the set of realizable spectral maps.

Lemma 3. Consider a graphical Horn problem of dimension d .

- The zero spectral map $0_d : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ assigning to $e \in \mathcal{E}$, the spectrum $0_d(e) = (0, \dots, 0)$ is realizable.
- If $\alpha : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^a$ and $\beta : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^b$ are realizable, then their direct sum $\alpha \oplus \beta : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^{a+b}$ and tensor product $\alpha \otimes \beta : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^{ab}$ (defined pointwise), are both realizable.
- If $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is a realizable and $p \in \mathbb{R}_{>0}$ a positive real number, then $p\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$, defined by $(p\lambda)(e) = (p\lambda_1(e), \dots, p\lambda_d(e))$ is also a realizable.
- If $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is a realizable, then $\lambda^* : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$, defined by a pointwise application of the negation-reversal involution, is also realizable.

Example 1 (The trigonal Horn problem). Perhaps the smallest non-trivial example of a graphical Horn problem is the one for which Horn originally conjectured a solution [Hor62], namely, what are the possible eigenvalues, (α, β, γ) , of a triple of $d \times d$ Hermitian matrices (A, B, C) satisfying the condition $A + B = C$? As a graphical Horn problem,

$$\Gamma = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix} \quad (29)$$

From the perspective of Problem 3, we refer to this question as the *trigonal Horn problem*.

Example 2 (Polygonal Horn problems). If the directed graph $\mathcal{G}$ consists of a single closed cycle, then the corresponding Horn problem will be referred to as a *polygonal Horn problem*; the polygonal Horn problem associated to an *oriented* cycle $(e_1, e_2, \dots, e_r, e_1)$ consisting of r edges asks what are the possible eigenvalues $(\lambda_1, \dots, \lambda_r)$ of r Hermitian matrices $(H_1, \dots, H_r)$ with zero-sum $H_1 + \dots + H_r = 0$.

$$\Gamma = \begin{pmatrix} 1 & \cdot & \cdot & \cdot & 1 \end{pmatrix} \quad (30)$$

Remark 3. Evidently, there is a bijection between the set of realizable spectra (α, β, γ) and the set of realizable spectra $(\lambda_1, \lambda_2, \lambda_3)$ for the polygonal Horn problem where $r = 3$: namely

$$(\lambda_1, \lambda_2, \lambda_3) = (\alpha, \beta, \gamma^*) \quad (31)$$

where $\gamma \mapsto \gamma^*$ is the involution on $\mathbb{R}_{\text{dec}}^d$ defined by negation of entries and then reversing to preserve the ordering:

$$\gamma^* = (-\gamma_d, \dots, -\lambda_1). \quad (32)$$

Example 3 (Tetrahedral Horn problem). In this paper, a considerable focus will be placed on the *tetrahedral Horn problem* which is the graphical Horn problem associated with the directed graph above forming a tetrahedron. Specifically, a sextuple of candidate spectra,

$$(\alpha, \beta, \gamma, \mu, \nu, \lambda) \in (\mathbb{R}_{\text{dec}}^d)^{\times 6} \quad (33)$$

are said to be *tetrahedral* if they are the respective eigenvalues of Hermitian matrices

$$(A, B, C, M, N, L) \in \mathcal{H}(d)^{\times 6} \quad (34)$$

satisfying the following four conditions:

$$A + B = M, \quad M + C = L, \quad B + C = N, \quad A + N = L. \quad (35)$$

As a graphical Horn problem,

$$\Gamma = \begin{pmatrix} 1 & 1 & 0 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & 1 & -1 \end{pmatrix} \quad (36)$$

Example 4 (One-dimensional graphical Horn problems). An important special case of graphical Horn problem for a digraph $\mathcal{G}$ is the case where $d = 1$. If $d = 1$, then vector space of 1×1 Hermitian matrices becomes isomorphic to the real numbers, $\mathcal{H}(1) \simeq \mathbb{R}$, and similarly the space of candidate eigenvalues becomes $\mathbb{R}_{\text{dec}}(1) = \mathbb{R}$. Moreover, the spectral map $\lambda : \mathcal{E} \rightarrow \mathbb{R}$ becomes an assignment of real numbers of the edges of $\mathcal{G}$, the map $H : \mathcal{E} \rightarrow \mathbb{R}$ must necessarily equal spectral map λ by the spectral condition Eq. (25), and the cycle condition characterizing realizable spectral maps simplifies to the condition that λ , viewed as a vector $(\lambda(e))_{e \in \mathcal{E}} \in \mathbb{R}^{|\mathcal{E}|}$, must lie in the kernel of $\Gamma(\mathcal{G})$,

$$\lambda \in \ker(\Gamma(\mathcal{G})). \quad \text{\textcolor{blue}{(eq:ker_cond_d_1)}} \quad (37)$$

In fact, this simple condition provides a complete solution to all graphical Horn problems where $d = 1$.

The solution to all one-dimensional graphical Horn problems provided by Eq. (37) also gives rise to a simple necessary condition for $d > 1$ graphical Horn problems. \textcolor{blue}{lem:trace_cond}

Lemma 4. Consider the graphical Horn problem of dimension d for the digraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Then the spectral map $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is realizable only if

$$\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G})), \quad (38)$$

where, by abuse of notation, $\text{Tr} : \mathbb{R}_{\text{dec}}^d \rightarrow \mathbb{R}$ denotes the sum

$$\text{Tr}((\lambda_1, \dots, \lambda_d)) := \lambda_1 + \dots + \lambda_d. \quad (39)$$

Proof. Suppose that $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is a spectral map realized by the assignment $H : \mathcal{E} \rightarrow \mathcal{H}(d)$, meaning $\text{eig}(H(e)) = \lambda(e)$ for all edges $e \in \mathcal{E}$, and $\sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} H(e) = 0$ for all cycles ℓ . Taking the trace of cycle condition Eq. (26) yields

$$0 = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(H(e)) = \sum_{e \in E} \Gamma(\mathcal{G})_{\ell, e} \text{Tr}(\lambda(e)). \quad (40)$$

In other words, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ which implies $\text{Tr} \circ \lambda$ is a realizable spectral map for the $d = 1$ case. $\square$

Lemma 5. Consider the graphical Horn problem of dimension d for the diagraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Suppose $\mu : \mathcal{E} \rightarrow \mathbb{R}$ lies in the kernel of $\Gamma(\mathcal{G})$ and define $\mu \cdot 1_d : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ to be the spectral map defined by

$$(\mu \cdot 1_d)(e) := (\mu(e), \dots, \mu(e)). \quad (41)$$

Then $\mu \cdot 1_d : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is a realizable spectral map. Moreover, $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is a realizable spectral map if and only if $\lambda + \mu \cdot 1_d$ is a realizable.

Proof. The spectral map $\mu \cdot 1_d$ is realized by the assignment $H_\mu : \mathcal{E} \rightarrow \mathcal{H}(d)$ defined by

$$H_\mu(e) := \mu(e)I_d \quad (42)$$

where I_d is the $d \times d$ identity matrix. This is because it was assumed that $\mu \in \ker(\Gamma(\mathcal{G}))$ and therefore

$$\sum_{e \in \mathcal{E}} \Gamma(\mathcal{G})_{\ell, e} H_\mu(e) = \left(\sum_{e \in \mathcal{E}} \Gamma(\mathcal{G})_{\ell, e} \mu(e) \right) I_d = 0. \quad (43)$$

Now suppose $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ is realized by the assignment $H_\lambda : \mathcal{E} \rightarrow \mathcal{H}(d)$, then it follows that $H_\lambda + H_\mu$, defined by

$$(H_\lambda + H_\mu)(e) = H_\lambda(e) + \mu(e)I_d \quad (44)$$

realizes the spectral map $\lambda + \mu \cdot 1_d$ because the spectrum of $H_\lambda(e) + \mu(e)I_d$ is simply $\lambda(e) + \mu \cdot 1_d(e)$ (the eigenvalues add because $[H_\lambda(e), I_d] = 0$). $\square$

A direct corollary of Lemma 5 is that the set of realizable spectral maps, $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$, is entirely determined by the set of realizable and *traceless* spectral maps in the sense that $\text{Tr} \circ \lambda = 0_d$.

Corollary 6. Let $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ be a spectral map such that $\text{Tr} \circ \lambda \in \ker(\Gamma(\mathcal{G}))$ and let $\lambda^0 : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ be the traceless spectral map defined by

$$\lambda^0(e) := (\lambda_1(e) - \text{Tr}(\lambda(e)), \dots, \lambda_d(e) - \text{Tr}(\lambda(e))). \quad (45)$$

Then λ is realizable if and only if λ^0 is realizable.

Proof. The spectral map λ^0 is obtained from λ by subtracting the spectral map $\text{Tr} \circ \lambda$. By assumption, $\text{Tr} \circ \lambda$ lies in the kernel of $\Gamma(\mathcal{G})$ by Lemma 4, and thus Lemma 5 (when $\mu \mapsto \pm \text{Tr} \circ \lambda$) implies λ^0 is realizable if and only if λ is realizable. $\square$

Corollary 7. Suppose $\mathcal{G}$ is a graphical Horn problem such that $\ker(\Gamma(\mathcal{G}))$ contains a one-dimensional spectral map $\rho : \mathcal{E} \rightarrow \mathbb{R}$ which is strictly positive, meaning

$$\forall e \in \mathcal{E} : \rho(e) > 0. \quad (46)$$

Let $\lambda : \mathcal{E} \rightarrow \mathbb{R}_{\text{dec}}^d$ be a spectral map. Then there exists a sufficiently large positive $p \in \mathbb{R}_{>0}$ such that $\lambda^+ := \lambda + p\rho \cdot 1_d$ is a spectral map consisting of strictly positive spectra meaning

$$\forall e \in \mathcal{E} : \lambda_d^+(e) > 0. \quad (47)$$

{eq:positive_spectra}

Proof. The value of $\lambda_d^+(e)$ for $e \in \mathcal{E}$ is determined by

$$\lambda_d^+(e) = \lambda_d(e) + p\rho(e). \quad (48)$$

In order to satisfy Eq. (47) it suffices to let p satisfy

$$p > \max_{e \in \mathcal{E}} \frac{-\lambda_d(e)}{\rho(e)}. \quad (49)$$

$\square$

Example 5. The tetrahedral Horn problem defined above admits of strictly positive one-dimensional spectral maps. Specifically, $\rho = (a, b, c, a + b, b + c, a + b + c)$ lies in the kernel of Γ and is strictly positive whenever $a, b, c > 0$.

TODO: Talk about subgraphs and projections, adhesivity proposition, etc.

Remark 4. It is also worth noting that the existence of a solution to a graphical Horn problem is eventually unchanging when considering the limit as $d \rightarrow \infty$. Specifically, if there exists a solution for a particular value of d , then there exists solutions for all $d' > d$ obtained by padding the Hermitian matrices with zeros. Moreover, if there exists a solution for any value of d , then there will necessarily exist a solution involving $d' \times d'$ matrices where $d' = \sum_e \text{rank}(e)$ is the sum of the ranks (number of non-zero values) of all the eigenvalue lists involved. TODO: What is the smallest value of d' which contains a solution is there is a solution for any d ? Consider the following example which demonstrates that the ranks of the eigenvalue lists can in principle be much smaller than the dimension d :

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \quad (50)$$

What about what happens when positivity conditions are also imposed?

3 A representation-theoretic characterization

The purpose of this section is to widen the bridge between representation theory and Horn-type problems. We will begin with a treatment of the connection between triangular spectra and the Littlewood-Richardson coefficients and discuss its connection to symplectic geometry and moment maps. Then we will articulate the manner in which tetrahedral spectra are related to the norm of the recoupling operator for $SU(d)$ and discuss the manner in which convexity results for moment maps cannot be applied and thus one must not expect the space of tetrahedral spectra to be convex (indeed it is not).

There is a well-established and fundamental connection between the eigenvalues of sums of Hermitian matrices and the representation theory of the general linear group [Kly98; KT99; Knu00; Ful00; Bha01; Chr08; BVW18]. The purpose of this section is to briefly summarize the nature of this connection and then describe the manner in which this fundamental connection generalizes to graphical Horn problems.

3.1 Representation & geometric invariant theory preliminaries

sec:prelim

There are a number of approaches that can be taken to begin building the bridge which connects eigenvalues of sums of $d \times d$ Hermitian matrices to the representation theory of $GL(d)$; the method taken here is through notions of stability in the sense of geometric invariant theory. We begin by recalling prerequisite notions from representation theory, symplectic geometry and geometric invariant theory, establishing our notation along the way.

As always, we identify the set of $d \times d$ Hermitian matrices, $\mathcal{H}(d)$ with the set $\mathfrak{u}(d)^*$ of linear functionals on the Lie algebra $\mathfrak{u}(d)$ of $U(d)$ using the non-degenerate bilinear pairing $(H, X) = i \operatorname{Tr}(XH)$ for $H \in \mathcal{H}(d)$ Hermitian and $X \in \mathfrak{u}(d)$ skew-Hermitian. Given $\lambda \in \mathbb{R}_{\text{dec}}^d$, let $D_\lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_d)$ be the diagonal $d \times d$ Hermitian matrix. The set $\mathcal{O}_\lambda$ of $d \times d$ Hermitian matrices with eigenvalues λ can be identified with the $U(d)$ -orbit of D_λ under the adjoint action, $U \cdot D_\lambda = UD_\lambda U^{-1}$, i.e.

$$\mathcal{O}_\lambda = U(d) \cdot D_\lambda = \{U \cdot D_\lambda \mid U \in U(d)\}, \quad (51)$$

or equivalently identified with a coadjoint orbit in $\mathfrak{u}(d)^*$. When $\lambda \in \mathbb{Z}_{\text{dec}}^d$ is integer-valued, it may additionally be considered as a dominant weight for $GL(d)$. In such cases, let $\pi_\lambda : GL(d) \rightarrow \operatorname{Aut}(V_\lambda)$ ¹ denote the unique (up to isomorphism) irreducible representation of $GL(d)$ with highest weight λ and let $v_\lambda \in V_\lambda$ be a highest weight vector and let $[v_\lambda] \in \mathbb{P}(V_\lambda)$ be the associated highest weight ray. One important consequence of v_λ being a highest weight vector is simply that the $B(d)$ -orbit of v_λ , $B(d) \cdot v_\lambda = \{\pi_\lambda(b)v_\lambda\}$, is contained within the ray $[v_\lambda]$. In addition, as each $g \in GL(d)$ admits of a decomposition, $g = ub$, for $(u, b) \in U(d) \times B(d)$, the $U(d)$ -orbit of $[v_\lambda]$ is equivalent to the $GL(d)$ -orbit of $[v_\lambda]$. Moreover, there is a well-defined $U(d)$ -equivariant diffeomorphism

$$\varphi : \mathcal{O}_\lambda \rightarrow U(d) \cdot [v_\lambda] \quad (52)$$

sending $u \cdot D_\lambda \in \mathcal{O}_\lambda$ to $u \cdot [v_\lambda] = [v_\lambda^u] \in U(d) \cdot [v_\lambda] \subseteq \mathbb{P}(V_\lambda)$ ² where

$$\varphi(u \cdot D_\lambda) := v_\lambda^u := \pi_\lambda(u)v_\lambda. \quad (53)$$

Furthermore, if $X \in \mathfrak{gl}(d)$ is a $d \times d$ matrix, then it can be shown that $\operatorname{Tr}(XD_\lambda) = \langle v_\lambda, \Pi_\lambda(X)v_\lambda \rangle$ where Π_λ is the induced representation of $\mathfrak{gl}(d)$ on V_λ , $\langle \cdot, \cdot \rangle$ is a $U(d)$ -invariant inner product on V_λ and v_λ is a highest weight vector of unit norm in V_λ . Using the aforementioned diffeomorphism, one can also show that for all $d \times d$ matrices $X \in \mathfrak{gl}(d)$,

$$\operatorname{Tr}(X(u \cdot D_\lambda)) = \langle v_\lambda^u, \Pi_\lambda(X)v_\lambda^u \rangle. \quad (54)$$

¹The representation space V_λ can be realized concretely in the sense of Borel-Weil as follows. Let $B(d)$ be the Borel subgroup of upper triangular matrices in $GL(d)$ and $F(d) = GL(d)/B(d)$ the corresponding flag variety/manifold with $GL(d) \rightarrow F(d)$ the principle $B(d)$ -bundle over $F(d)$. Given a weight $\mu \in \mathbb{Z}^d$, let χ_μ be the one-dimensional complex representation of $B(d)$ sending the invertible diagonal $D_t = \operatorname{diag}(t_1, \dots, t_d)$ to $\chi_\mu(D_t) = t_1^{\mu_1} \dots t_d^{\mu_d} \in \mathbb{C}$. The representation χ_μ of $B(d)$ gives rise to the associated line bundle, $GL(d) \times_{B(d)} \mathbb{C} \rightarrow F(d)$. With respect to this construction, the representation space V_λ can be identified with the vector space of holomorphic sections of the associated line bundle given by $\chi_{-\lambda^*}$, i.e. holomorphic functions $s : GL(d) \rightarrow \mathbb{C}$ satisfying $s(gb) = s(g)\chi_{\lambda^*}(b)$ for all $g \in GL(d)$ and $b \in B(d)$. **TODO: Check to signs in this footnote to make sure they are correct.**

²This map is well-defined because the $U(d)$ -stabilizers of D_λ and of v_λ are equivalent.

Remark 5 (Highest and lowest weights). Let $\lambda = (\lambda_1, \dots, \lambda_d)$ be a dominant weight for $\mathrm{GL}(d)$, and V_λ the unique up to isomorphism finite dimensional irreducible representation of $\mathrm{GL}(d)$ with highest weight λ and unique (up to phase) highest weight vector $v_\lambda \in V_\lambda$. There exists a unique $\mathrm{U}(d)$ -invariant inner product $\langle \cdot, \cdot \rangle$ on V_λ which enables one to identify elements, $f : V_\lambda \rightarrow \mathbb{C}$, in the dual vector space, V_λ^* , with elements $w \in V_\lambda$ with the original vector space such that $f(v) = \langle w, v \rangle$ holds for all $v \in V_\lambda$. Moreover, the dual representation of $\mathrm{GL}(d)$ satisfies

$$[\pi_\lambda^*(g)f](v) = \langle w, \pi_\lambda(g^{-1})v \rangle. \quad (55)$$

The highest weight appearing in the dual representation, V_λ^* , is the dominant weight

$$\lambda^* = (-\lambda_d, \dots, -\lambda_1). \quad (56)$$

Consequently, there exists an $\mathrm{GL}(d)$ -equivariant isomorphism $V_{\lambda^*} \simeq V_\lambda^*$, justifying the abuse of notation. Under this isomorphism, the vector $\langle v_\lambda, \cdot \rangle \in V_\lambda^*$ dual to $v_\lambda \in V_\lambda$ is a vector of *lowest* weight in V_{λ^*} . To obtain the vector v_{λ^*} of *highest* weight in V_{λ^*} , one must dualize the vector $v_{w_0(\lambda)} := \pi_\lambda(w_0)v_\lambda$ of *lowest* weight in V_λ , i.e.,

$$v_{\lambda^*} \simeq \langle \pi_\lambda(w_0)v_\lambda, \cdot \rangle \quad (57)$$

where $w_0 \in S_d$ is the permutation matrix which reverses the order of elements sending $(\lambda_1, \dots, \lambda_d)$ to $(\lambda_d, \dots, \lambda_1)$.

This notational convention is consistent with the diffeomorphism $\varphi : \mathcal{O}_\lambda \rightarrow \mathrm{U}(d) \cdot [v_\lambda]$ defined above in the following sense; if $X = u \cdot D_\lambda = u D_\lambda u^{-1}$ is a Hermitian matrix with eigenvalues equal to the dominant weight λ , then $-X = (uw_0) \cdot D_{\lambda^*}$ is a Hermitian matrix with eigenvalues equal to the dominant weight $\lambda^* = -w_0(\lambda)$. Moreover the dual of $\varphi(X) = \pi_\lambda(u)v_\lambda$ under the inner product on V_λ becomes (under the $V_{\lambda^*} \simeq V_\lambda^*$ isomorphism)

$$\varphi(-X) = \pi_{\lambda^*}(uw_0)v_{\lambda^*} \simeq \langle \pi_\lambda(w_0)v_\lambda, \pi_\lambda((uw_0)^{-1}) \cdot \rangle = \langle \pi_\lambda(u)v_\lambda, \cdot \rangle = \langle \varphi(X), \cdot \rangle. \quad (58)$$

Throughout this section, the following result serves as the basis for establishing a number of asymptotic results for large k .

lem:kth_tensor_power

Lemma 8. *Let $\lambda \in \mathbb{Z}_{\mathrm{dec}}^d$ be a dominant weight for $\mathrm{GL}(d)$, let V_λ be the irreducible representation of $\mathrm{GL}(d)$ with highest weight λ and let $k \in \mathbb{N}_{>0}$ be a positive integer. Then $\dim(\mathrm{Hom}_{\mathrm{GL}(d)}(V_{k\lambda}, V_\lambda^{\otimes k})) = 1$. Let*

$$\iota_\lambda^k : V_{k\lambda} \rightarrow V_\lambda^{\otimes k} \quad (59)$$

be a $\mathrm{GL}(d)$ -intertwining isometry in $\mathrm{Hom}_{\mathrm{GL}(d)}(V_{k\lambda}, V_\lambda^{\otimes k})$. If $v_\lambda \in V_\lambda$ is a highest weight vector, then there exists a highest weight vector $v_{k\lambda} \in V_{k\lambda}$ satisfying

$$\iota_\lambda^k v_{k\lambda} = v_\lambda^{\otimes k}. \quad (60)$$

Moreover, for all $g \in \mathrm{GL}(d)$,

$$\|\pi_{k\lambda}(g)v_{k\lambda}\| = \|\pi_\lambda(g)v_\lambda\|^k. \quad (61)$$

Proof. TODO: The following proof only makes sense for $\mathrm{SL}(d)$ because I'm implicitly assuming the labelling dominant weights are taken to be Young diagrams. Fix the proof so that it holds for $\mathrm{GL}(d)$. That $\dim(\mathrm{Hom}_{\mathrm{GL}(d)}(V_{k\lambda}, V_\lambda^{\otimes k})) = 1$ follows from

$$c_{\lambda, (k-1)\lambda}^{k\lambda} = \dim(\mathrm{Hom}_{\mathrm{GL}(d)}(V_{k\lambda}, V_\lambda^{\otimes k})) \quad (62)$$

and the Littlewood-Richardson rule applied to $c_{\lambda, (k-1)\lambda}^{k\lambda}$; the skew shape $k\lambda/(k-1)\lambda$ has λ_j boxes appearing in row j and consequently there is only one skew semistandard tableaux of shape $k\lambda/(k-1)\lambda$ with weight λ (and moreover this tableaux is indeed a Littlewood-Richardson tableaux so $c_{\lambda, (k-1)\lambda}^{k\lambda} = 1$). The existence of $v_{k\lambda}$ follows from the fact that $v_\lambda^{\otimes k}$ is already a highest weight vector of weight $k\lambda$ and therefore necessarily lives in image of ι_λ^k . Finally, let $g \in \mathrm{GL}(d)$ be arbitrary. Then

$$\|\pi_{k\lambda}(g)v_{k\lambda}\| = \|\iota_\lambda^k \pi_{k\lambda}(g)v_{k\lambda}\| = \|\pi_\lambda(g)^{\otimes k} \iota_\lambda^k v_{k\lambda}\| = \|\pi_\lambda(g)^{\otimes k} v_\lambda^{\otimes k}\| = \|\pi_\lambda(g)v_\lambda\|^k. \quad (63)$$

□

3.2 Reformulating the cycle condition

[sec:reform_cycle](#)

It will be shown how the cycle condition, Eq. (26), is equivalent to the existence of certain vectors which are minimal in norm with respect to a representation of $\mathrm{GL}(d)$. Now consider a polygonal graph consisting of a single connected cycle of length s such that Γ is a $\{\pm 1\}$ -valued $1 \times s$ matrix. Let $\lambda : [s] \rightarrow \mathbb{R}_{\mathrm{dec}}^d$ be an eigenvalue map for such a Γ , we consider the symplectic manifold

$$\mathcal{M}_\ell := \prod_{e \in \mathcal{E}} \mathcal{O}_{\lambda(e)\gamma_\ell(e)} \quad (64)$$

It can be shown that the moment map $\Phi_\ell : \mathcal{M}_\ell \rightarrow \mathcal{H}(d)$ associated to this action satisfies

$$\Phi_\ell(\gamma_\ell(e)H(e) \mid e \in \ell) = \sum_{e \in \ell} \gamma_\ell(e)H(e) \quad (65)$$

and thus vanishes precisely when the cycle condition, Eq. (26), is satisfied.

3.3 Littlewood-Richardson coefficients & recoupling operators

[defn:lr_coeffs](#)

Definition 2 (Littlewood-Richardson coefficients). Let $\alpha, \beta, \gamma \in \mathbb{Z}_{\mathrm{dec}}^d$ be a triple of dominant weights for $\mathrm{GL}(d)$ and let $V_\alpha, V_\beta, V_\gamma$ be the triple of irreducible representations of $\mathrm{GL}(d)$ with corresponding highest weights. Let $C_{\alpha, \beta}^\gamma$ denote the multiplicity space of $\mathrm{GL}(d)$ -intertwiners from V_γ to $V_\alpha \otimes V_\beta$:

$$C_{\alpha, \beta}^\gamma := \mathrm{Hom}_{\mathrm{GL}(d)}(V_\gamma, V_\alpha \otimes V_\beta) \simeq \mathrm{Inv}_{\mathrm{GL}(d)}(V_{\gamma^*} \otimes V_\alpha \otimes V_\beta). \quad (66)$$

The *Littlewood-Richardson coefficient* is defined as the dimension of this multiplicity space:

$$c_{\alpha, \beta}^\gamma := \dim(C_{\alpha, \beta}^\gamma). \quad (67)$$

Definition 3 (Clebsch-Gordon isometries). Given α, β, γ as in Definition 2, a *Clebsch-Gordon isometry* is a $\mathrm{GL}(d)$ -equivariant isometric linear map sending $f \otimes v \in C_{\alpha, \beta}^\gamma \otimes V_\gamma$ to $f(v) \in V_\alpha \otimes V_\beta$ ³. Let

$$Y_{\alpha, \beta}^\gamma : C_{\alpha, \beta}^\gamma \otimes V_\gamma \rightarrow V_\alpha \otimes V_\beta, \quad (68)$$

be a Clebsch-Gordon isometry and depict it diagrammatically by

$$\begin{array}{c} V_\alpha \\ \uparrow \\ \text{---} \circ \text{---} \\ \uparrow \quad \uparrow \\ C_{\alpha, \beta}^\gamma \quad V_\gamma \end{array} = \begin{array}{c} V_\alpha \quad V_\beta \\ \uparrow \quad \uparrow \\ \boxed{Y_{\alpha, \beta}^\gamma} \\ \uparrow \quad \uparrow \\ C_{\alpha, \beta}^\gamma \quad V_\gamma \end{array}. \quad (69)$$

The conjugate transpose of the Clebsch-Gordon isometry is

$$Y_{\alpha, \beta}^{\gamma\dagger} : V_\alpha \otimes V_\beta \rightarrow C_{\alpha, \beta}^\gamma \otimes V_\gamma, \quad (70)$$

will be depicted diagrammatically by

$$\begin{array}{c} C_{\alpha, \beta}^\gamma \quad V_\gamma \\ \uparrow \quad \uparrow \\ \text{---} \circ \text{---} \\ \uparrow \quad \uparrow \\ V_\alpha \quad V_\beta \end{array} = \begin{array}{c} C_{\alpha, \beta}^\gamma \quad V_\gamma \\ \uparrow \quad \uparrow \\ \boxed{Y_{\alpha, \beta}^{\gamma\dagger}} \\ \uparrow \quad \uparrow \\ V_\alpha \quad V_\beta \end{array}. \quad (71)$$

[defn:ud_recoupling_operator](#)

Definition 4 (Recoupling operators). Let x, y, z and d be positive integers and let

$$\alpha \in \Upsilon_{x, d}, \delta \in \Upsilon_{z, d}, \beta \in \Upsilon_{y, d}, \epsilon \in \Upsilon_{x+y+z, d}, \gamma \in \Upsilon_{x+y, d}, \quad \phi \in \Upsilon_{y+z, d}. \quad (72)$$

³If $C_{\alpha, \beta}^\gamma = 0$, then $Y_{\alpha, \beta}^\gamma$ is identified with the origin in $V_\alpha \otimes V_\beta$.

The $U(d)$ -recoupling operator is the operator

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha,\beta}^\gamma \otimes C_{\delta,\gamma}^\epsilon \rightarrow C_{\alpha,\phi}^\epsilon \otimes C_{\beta,\delta}^\phi, \quad (73)$$

defined by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{\text{Tr}_{V_\epsilon}[Y_{\alpha,\phi}^{\epsilon\dagger}(\text{id}_{V_\alpha} \otimes Y_{\beta,\delta}^{\phi\dagger})(Y_{\alpha,\beta}^\gamma \otimes \text{id}_{V_\delta})Y_{\gamma,\delta}^\epsilon]}{\dim(V_\epsilon)}, \quad (74)$$

which is depicted diagrammatically by

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} := \frac{1}{(V_\epsilon)} \left[\begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \uparrow \\ Y_{\alpha,\phi}^{\epsilon\dagger} \\ \downarrow V_\phi \\ Y_{\beta,\delta}^{\phi\dagger} \\ \downarrow V_\beta \\ Y_{\alpha,\beta}^\gamma \\ \downarrow V_\gamma \\ Y_{\gamma,\delta}^\epsilon \\ \downarrow V_\delta \\ C_{\gamma,\delta}^\epsilon \end{array} \right] = \frac{1}{(V_\epsilon)} \left[\begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \uparrow \\ V_\alpha \\ \downarrow \\ C_{\alpha,\beta}^\gamma \end{array} \right] \left[\begin{array}{c} C_{\beta,\delta}^\phi \\ \uparrow \\ V_\beta \\ \downarrow \\ C_{\gamma,\delta}^\epsilon \end{array} \right] \left[\begin{array}{c} C_{\gamma,\delta}^\epsilon \\ \uparrow \\ V_\gamma \\ \downarrow \\ C_{\alpha,\beta}^\gamma \end{array} \right] \left[\begin{array}{c} C_{\alpha,\phi}^\epsilon \\ \uparrow \\ V_\alpha \\ \downarrow \\ C_{\beta,\delta}^\phi \end{array} \right]. \quad (75)$$

WARNING: The convention I am using for the curly braces seems to differ from many of the uses of the curly braces to define the Wigner 6j coefficient which differs in that I think the Wigner 6j coefficient does not include the additional factor of $\dim(V_\epsilon)$.

Subsequent sections will primarily be interested in the square of the Hilbert-Schmidt norm of the recoupling operator in Definition 4. While the individual coefficients of the recoupling operator are dependent on a choice of basis for each of the multiplicity spaces, its Hilbert-Schmidt and operator norms are not dependent on such a choice of basis. The following lemma provides a formula for the square of the Hilbert-Schmidt norm as an integral over $SU(d)^{\times 4}$.

lem:character_formula

Lemma 9 (Character formula). *Let $\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha,\beta}^\mu \otimes C_{\mu,\gamma}^\lambda \rightarrow C_{\alpha,\nu}^\lambda \otimes C_{\beta,\gamma}^\nu$ be the recoupling operator defined in Definition 4. Then*

$$R_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma} := \dim(V_\epsilon)^2 \left\| \left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} \right\|_{\text{HS}}^2 = \int_{SU(d)^4} du_1 du_2 du_3 du_4 X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4). \quad (76)$$

feg:character_formula

where

$$X_{\delta,\epsilon,\phi}^{\alpha,\beta,\gamma}(u_1, u_2, u_3, u_4) := \chi_\alpha(u_1^{-1}u_2)\chi_\beta(u_4^{-1}u_2)\chi_\gamma(u_2^{-1}u_3)\chi_\delta(u_4^{-1}u_3)\chi_\epsilon(u_3^{-1}u_1)\chi_\phi(u_1^{-1}u_4) \quad (77)$$

Note that when $d = 2$, Eq. (76) recovers the well-known formula for the square of the 6j-symbol for $SU(2)$ (see [FL03, Equation (1)] and also [Bar98; FK00]).

Proof of Lemma 9. The proof follows from (i) the following identities concerning the character $\chi_\lambda(u) = \text{Tr}(\pi_\lambda(u))$ and duality $\pi_\lambda^*(u) = (\pi_\lambda(u^{-1}))^T$ relations

$$\chi_\lambda(u) = \boxed{\pi_\lambda(u)} \begin{array}{c} \uparrow \lambda \\ \downarrow \lambda \end{array} \quad \pi_\lambda^*(u) = \boxed{\pi_\lambda^*(u)} \begin{array}{c} \uparrow \gamma \\ \downarrow \gamma \end{array} = \boxed{\pi_\gamma(u^{-1})} \begin{array}{c} \uparrow \gamma \\ \downarrow \gamma \end{array}, \quad (78)$$

and (ii) the integral formula for the projection operator $\Pi_{\alpha,\beta}^\gamma$ onto the subspace of invariant vectors in $V_\alpha \otimes V_\beta \otimes V_\gamma^*$:

$$\begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\Pi_{\alpha,\beta}^\gamma} \\ \uparrow \quad \uparrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array} = \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \\ C_{\alpha,\beta}^\gamma \\ \curvearrowleft \quad \curvearrowleft \quad \curvearrowleft \\ \alpha \quad \beta \quad \gamma \end{array} = \int du \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \uparrow \quad \uparrow \quad \downarrow \\ \boxed{\pi_\alpha(u)} \quad \boxed{\pi_\beta(u)} \quad \boxed{\pi_\gamma^*(u)} \\ \uparrow \quad \uparrow \quad \downarrow \\ \alpha \quad \beta \quad \gamma \end{array}. \quad (79)$$

Applied four times, we obtain

$$\begin{aligned} & \left(\bigcirc_{\epsilon}^2 C_{\alpha,\phi}^\epsilon \right) C_{\alpha,\beta}^\gamma C_{\gamma,\delta}^\epsilon C_{\beta,\delta}^\phi = \begin{array}{c} \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \\ \downarrow C_{\alpha,\phi}^\epsilon \quad \downarrow C_{\alpha,\beta}^\gamma \quad \downarrow C_{\gamma,\delta}^\epsilon \quad \downarrow C_{\beta,\delta}^\phi \\ \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \end{array} \\ &= \Pi_i \int du_i \begin{array}{c} \bar{u}_1 \quad u_1 \quad \bar{u}_1 \quad u_2 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_3 \quad u_4 \quad \bar{u}_4 \quad u_4 \\ \phi \quad \epsilon \quad \alpha \quad \alpha \quad \beta \quad \gamma \quad \gamma \quad \delta \quad \epsilon \quad \delta \quad \beta \quad \phi \end{array} \\ &= \Pi_i \int du_i \begin{array}{c} \bar{u}_1 \quad u_2 \quad \bar{u}_2 \quad u_3 \quad u_3 \quad \bar{u}_4 \quad u_2 \quad \bar{u}_4 \quad u_1 \quad \bar{u}_3 \quad \bar{u}_1 \quad u_4 \\ \alpha \quad \gamma \quad \delta \quad \beta \quad \epsilon \quad \phi \end{array}. \quad (80) \end{aligned}$$

□

Remark 6. With the integral character formula $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ established, it is interesting to compare it against the character formula for the heat kernel of a compact Lie group G [Feg78], denoted by $K_G(g_1, g_2; t)$ in [McS21]:

$$K_G(g_1, g_2; t) = \sum_{\lambda} \dim V_{\lambda}^d \chi_{\lambda}(g_1 g_2^{-1}) e^{-t(|\lambda+\rho|^2 - |\rho|^2)/2}, \quad (81)$$

where the sum is over all dominant integral weights λ for G and ρ the associated Weyl vector. It is conceivable that there might be a method for expressing $R_{\delta\epsilon\phi}^{\alpha\beta\gamma}$ as a low temperature limit, $t \rightarrow 0$, of an integral over products of heat kernels.

3.4 Trigonal & polygonal spectra

[sec:trigonal_spectra](#)

Our first result can be viewed as a simple condition which is sufficient to conclude that a given triple of integer eigenvalues are trigonal.

Lemma 10. *If $c_{\alpha,\beta}^\gamma > 0$, then there exists Hermitian matrices $A \in \mathcal{O}_\alpha$, $B \in \mathcal{O}_\beta$, and $C \in \mathcal{O}_\gamma$, satisfying $A + B - C = 0$.*

Proof. Our proof proceeds in the style of [BVW18, Proposition 2.3]. Given any dominant weight $\lambda \in \mathbb{Z}_{\text{dec}}^d$, let P_λ denote the Hermitian projection operator onto the invariant $\square$

Using the theory of spectrum estimation developed in the context of quantum information theory [ARS88; KW01; HM02], the following pair of theorems were proven in [Chr08]. Recall that $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^d)^{\times 3}$ are said to be *trigonal* if there exists Hermitian matrices $A, B, C \in \mathcal{H}(d)$ satisfying (i) $A + B + C = 0$ and (ii)

$$(\text{eig}(A), \text{eig}(B), \text{eig}(C)) = (\alpha, \beta, \gamma). \quad (82)$$

Theorem 11. *If $c_{\alpha, \beta, \gamma} > 0$ for some $\alpha, \beta, \gamma \in \mathbb{Z}_{\text{dec}}^d$, then (α, β, γ) are trigonal.*

Theorem 12. *If $(\alpha, \beta, \gamma) \in (\mathbb{R}_{\text{dec}}^d)^{\times 3}$ are triangular, then there exists a graded sequence $(\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)) \in (\mathbb{Z}_{\text{dec}}^d)^{\times 3}$ such that (i) for all $n \in \mathbb{N}_{>0}$,*

$$c_{\tilde{\alpha}(n), \tilde{\beta}(n), \tilde{\gamma}(n)} > 0, \quad (83)$$

and (ii)

$$\lim_{n \rightarrow \infty} \frac{\tilde{\alpha}(n)}{n} = \alpha, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\beta}(n)}{n} = \beta, \quad \lim_{n \rightarrow \infty} \frac{\tilde{\gamma}(n)}{n} = \gamma. \quad (84)$$

prop:kempf-ness-abc

Proposition 13 (Kempf-Ness). *Let $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ be a triple of $d \times d$ Hermitian matrices with integer-valued eigenvalues and let*

$$w_{A, B, C} := \varphi(A) \otimes \varphi(B) \otimes \varphi(C) = v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C} \quad (85)$$

be the associated coherent state of unit norm in $V_\alpha \otimes V_\beta \otimes V_\gamma$. Define the capacity of (A, B, C) as

$$\text{cap}(A, B, C) := \inf_{g \in \text{GL}(d)} \left\| [\pi_\alpha(g) \otimes \pi_\beta(g) \otimes \pi_\gamma(g)] w_{A, B, C} \right\|^2, \quad (86)$$

$$= \inf_{g \in \text{GL}(d)} \left\| v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C} \right\|^2. \quad (87)$$

Then $\text{cap}(A, B, C) \in [0, 1]$ with

$$\text{cap}(A, B, C) = 1 \iff \mu(A, B, C) = A + B + C = 0. \quad (88)$$

Proof. Summarize the direct and pedagogical Kempf-Ness style proof in [BVW18, Proposition 2.3]. $\square$

The following is a combination of [FW20, Theorem 1 & Proposition 3.1] (using the convention where capacities and constant portions are defined in terms of squared norms instead of norms directly).

Theorem 14 (Strong-duality & Occasionality). *Let everything be as in Proposition 13 and let k be a positive integer. Define the constant portion of A, B, C of degree k to be*

$$\text{cst}_k(A, B, C) = \left\| \Gamma_{\alpha, \beta, \gamma}^k w_{A, B, C}^{\otimes k} \right\|^{\frac{2}{k}}, \quad (89)$$

where $\Gamma_{\alpha, \beta, \gamma}^k$ is the projection operator onto the subspace of $\text{GL}(d)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. Then

$$\text{cap}(A, B, C) = \sup_k \text{cst}_k(A, B, C) \quad (90)$$

{eq:strong-duality}

Moreover, when $\text{cap}(A, B, C) = 1$ and $\mu(A, B, C) = 0$, there exists an integer $m > 0$ such that

$$\lim_{k \rightarrow \infty} (mk)^s \text{cst}_{mk}(A, B, C) > 0 \quad (91)$$

{eq:occasionality}

where $s = \dim U(d) - \dim U(d)_{[w_{A, B, C}]}$.

Corollary 15. Consider the quantity $\tilde{c}_{\alpha,\beta,\gamma}$ which measures the fraction of vectors $V_\alpha \otimes V_\beta \otimes V_\gamma$ which are $\text{GL}(d)$ -invariant:

$$\tilde{c}_{\alpha,\beta,\gamma} = \frac{\dim[(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\text{GL}(d)}]}{\dim(V_\alpha \otimes V_\beta \otimes V_\gamma)}. \quad (92)$$

If there exists $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then there exists an integer $K > 0$ such that $c_{K\alpha, K\beta, K\gamma} \geq 1^4$, and therefore for all integers $M > 0$

$$\tilde{c}_{MK\alpha, MK\beta, MK\gamma} \geq \frac{1}{\dim(V_{MK\alpha} \otimes V_{MK\beta} \otimes V_{MK\gamma})}. \quad (93)$$

On the other hand, if there does not exist $(A, B, C) \in \mathcal{O}_\alpha \times \mathcal{O}_\beta \times \mathcal{O}_\gamma$ such that $A + B + C = 0$, then the function $k \mapsto \tilde{c}_{k\alpha, k\beta, k\gamma}$ admits an exponential upper bound⁵

$$\tilde{c}_{k\alpha, k\beta, k\gamma} \leq \max_{A, B, C} \text{cap}(A, B, C)^k. \quad (94)$$

Proof. The lower-bound follows from the operator inequality

$$|v_\lambda^u\rangle \langle v_\lambda^u| \leq \text{id}_{V_\lambda}, \quad \text{\textcolor{blue}{\{eq:inclusivity\}}} \quad (95)$$

for all unit vectors $v_\lambda^u \in V_\lambda$ while the upper-bound follows from the resolution of identity over the normalized Haar measure du for $U(d)$:

$$\text{id}_{V_\lambda} = \dim(V_\lambda) \int_{u \in U(d)} du |v_\lambda^u\rangle \langle v_\lambda^u|. \quad \text{\textcolor{blue}{\{eq:resolution\}}} \quad (96)$$

Applying Eq. (96) three times to the definition of $\tilde{c}_{k\alpha, k\beta, k\gamma}$ yields

$$\begin{aligned} \tilde{c}_{k\alpha, k\beta, k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(d)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \\ &= \int du_A du_B du_C \langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha, k\beta, k\gamma}^1(v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle. \end{aligned} \quad \text{\textcolor{blue}{\{eq:tildc_k_integral\}}} \quad (97)$$

While the operator $\Gamma_{k\alpha, k\beta, k\gamma}^1$ appearing in Eq. (97) above projects onto the subspace of $\text{GL}(d)$ -invariant vectors in $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$, the operator $\Gamma_{\alpha, \beta, \gamma}^k$ projects onto the subspace of $\text{GL}(d)$ -invariant vectors in $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$. By Lemma 8, $V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma}$ is a subspace of $(V_\alpha \otimes V_\beta \otimes V_\gamma)^{\otimes k}$ with respect to a $\text{GL}(d)$ -equivariant isometry and therefore the projection operators satisfy

$$\Gamma_{k\alpha, k\beta, k\gamma}^1 = (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k)^\dagger \Gamma_{\alpha, \beta, \gamma}^k (\iota_\alpha^k \otimes \iota_\beta^k \otimes \iota_\gamma^k). \quad (98)$$

Moreover, using the $\text{GL}(d)$ -equivariance of $\Gamma_{\alpha, \beta, \gamma}^k$ and the inequality $\Gamma_{\alpha, \beta, \gamma}^k \leq (\text{id}_{V_\alpha \otimes V_\beta \otimes V_\gamma})^{\otimes k}$ yields, for all $g \in \text{GL}(d)$, an upper bound on the integrand in Eq. (97):

$$\langle (v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}), \Gamma_{k\alpha, k\beta, k\gamma}^1(v_{k\alpha}^{u_A} \otimes v_{k\beta}^{u_B} \otimes v_{k\gamma}^{u_C}) \rangle \quad (99)$$

$$= \langle (v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k}, \Gamma_{\alpha, \beta, \gamma}^k(v_\alpha^{u_A} \otimes v_\beta^{u_B} \otimes v_\gamma^{u_C})^{\otimes k} \rangle, \quad (100)$$

$$= \langle (v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C})^{\otimes k}, \Gamma_{\alpha, \beta, \gamma}^k(v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C})^{\otimes k} \rangle, \quad (101)$$

$$\leq \langle (v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C})^{\otimes k}, (v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C})^{\otimes k} \rangle, \quad (102)$$

$$= \left\| v_\alpha^{g u_A} \otimes v_\beta^{g u_B} \otimes v_\gamma^{g u_C} \right\|^{2k}. \quad (103)$$

Finally, let (u'_A, u'_B, u'_C) be unitaries which achieve the maximum of the integrand appearing in Eq. (97) (continuity of the integrand and compactness of $U(d)$ guarantees existence) such that $\tilde{c}_{k\alpha, k\beta, k\gamma}$ admits the bound

$$\tilde{c}_{k\alpha, k\beta, k\gamma} \leq \left\| v_\alpha^{g u'_A} \otimes v_\beta^{g u'_B} \otimes v_\gamma^{g u'_C} \right\|^{2k}. \quad (104)$$

⁴In fact the proof of the saturation conjecture implies $K = 1$ always suffices.

⁵The semigroup property of Littlewood-Richardson coefficients together with the at most polynomial growth in $\dim(V_{k\lambda})$ combined with the exponential upper bound on $\tilde{c}_{k\alpha, k\beta, k\gamma}$ actually implies $c_{\alpha, \beta, \gamma} = 0$ whenever $A + B + C = 0$.

Since $g \in \text{GL}(d)$ is arbitrary and (u'_A, u'_B, u'_C) is independent of g by definition one obtains

$$\tilde{c}_{k\alpha, k\beta, k\gamma} \leq \inf_{g \in \text{GL}(d)} \left\| v_{\alpha}^{gu'_A} \otimes v_{\beta}^{gu'_B} \otimes v_{\gamma}^{gu'_C} \right\|^{2k} = \text{cap}(A', B', C')^k \leq \max_{A, B, C} \text{cap}(A, B, C)^k. \quad (105)$$

TODO: Prove that the capacity is attained by some Hermitian matrices A, B, C (maybe using the quantitative Kempf-Ness theorem [Bür+19, Theorem 1.17]). This establishes the claimed upper bound. To prove the claimed lower bound, let $v_{\alpha}^{u'_A} \otimes v_{\beta}^{u'_B} \otimes v_{\gamma}^{u'_C}$ be the coherent state associated with the Hermitian matrices (A, B, C) satisfying $A + B + C = 0$. Then using Eq. (95) and the positive demidefiniteness of $\Gamma_{k\alpha, k\beta, k\gamma}^1$ we obtain,

$$\begin{aligned} \tilde{c}_{k\alpha, k\beta, k\gamma} &= \frac{\dim[(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})^{\text{GL}(d)}]}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})} \\ &\geq \frac{\langle (v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}), \Gamma_{k\alpha, k\beta, k\gamma}^1(v_{k\alpha}^{u'_A} \otimes v_{k\beta}^{u'_B} \otimes v_{k\gamma}^{u'_C}) \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \quad \{\text{eq:tilde_c_k_inclusivity}\} \\ &= \frac{\langle (v_{\alpha}^{u'_A} \otimes v_{\beta}^{u'_B} \otimes v_{\gamma}^{u'_C})^{\otimes k}, \Gamma_{\alpha, \beta, \gamma}^k(v_{\alpha}^{u'_A} \otimes v_{\beta}^{u'_B} \otimes v_{\gamma}^{u'_C})^{\otimes k} \rangle}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}, \\ &= \frac{\text{cst}_k(A, B, C)}{\dim(V_{k\alpha} \otimes V_{k\beta} \otimes V_{k\gamma})}. \end{aligned} \quad (106)$$

Now Eq. (91) implies there exists positive integers $m, s > 0$ such that for sufficiently large k , $(mk)^s \text{cst}_{mk}(A, B, C) > 0$ which implies $c_{mk\alpha, mk\beta, mk\gamma} \neq 0$ and thus $c_{mk\alpha, mk\beta, mk\gamma} \geq 1$. Taking $K = mk$ for sufficiently large k proves the claim. $\square$

3.5 Tetrahedral spectra & recoupling operators

[sec:tetraspec](#)

3.6 [Challenges with generalizing the scaling approach]

TODO: Show how to generalize the construction due to Roberts [Rob99] by taking the Cartesian product of these manifolds over all cycles and then imposing additional group actions and moment-map conditions to stitch everything together.

Remark 7. Consider the following vector space which is twelve-fold tensor product of irreducible representations of $\text{GL}(d)$:

$$\mathcal{L} = \mathcal{L}_{\delta\epsilon\phi}^{\alpha\beta\gamma} := V_{\alpha} \otimes V_{\alpha^*} \otimes V_{\beta} \otimes V_{\beta^*} \otimes V_{\gamma} \otimes V_{\gamma^*} \otimes V_{\delta} \otimes V_{\delta^*} \otimes V_{\epsilon} \otimes V_{\epsilon^*} \otimes V_{\phi} \otimes V_{\phi^*}. \quad (107)$$

along with an associated symplectic manifold

$$\mathcal{M} = \mathcal{M}_{\delta\epsilon\phi}^{\alpha\beta\gamma} := \mathcal{O}_{\alpha} \times \mathcal{O}_{\alpha^*} \times \mathcal{O}_{\beta} \times \mathcal{O}_{\beta^*} \times \mathcal{O}_{\gamma} \times \mathcal{O}_{\gamma^*} \times \mathcal{O}_{\delta} \times \mathcal{O}_{\delta^*} \times \mathcal{O}_{\epsilon} \times \mathcal{O}_{\epsilon^*} \times \mathcal{O}_{\phi} \times \mathcal{O}_{\phi^*}. \quad (108)$$

While the vector space $\mathcal{L}$ naturally supports a representation of $\text{GL}(d)^{\times 12}$, for the purposes of the tetrahedral Horn problem, there are three additional representations acting on $\mathcal{L}$ that will be of interest; $\sigma_6 : \text{GL}(d)^{\times 6} \rightarrow \text{GL}(\mathcal{L})$, $\sigma_4 : \text{GL}(d)^{\times 4} \rightarrow \text{GL}(\mathcal{L})$ and $\sigma_1 : \text{GL}(d) \rightarrow \text{GL}(\mathcal{L})$:

$$\begin{aligned} \sigma_6(g_1, g_2, g_3, g_4, g_5, g_6) &= \pi_{\alpha}(g_1) \otimes \pi_{\alpha^*}(g_1) \otimes \pi_{\beta}(g_2) \otimes \pi_{\beta^*}(g_2) \otimes \pi_{\gamma}(g_3) \otimes \pi_{\gamma^*}(g_3) \otimes \pi_{\delta}(g_4) \otimes \pi_{\delta^*}(g_4) \otimes \pi_{\epsilon}(g_5) \otimes \pi_{\epsilon^*}(g_5) \otimes \pi_{\phi}(g_6) \otimes \pi_{\phi^*}(g_6) \\ &\quad \cdots \otimes \pi_{\delta}(g_4) \otimes \pi_{\delta^*}(g_4) \otimes \pi_{\epsilon}(g_5) \otimes \pi_{\epsilon^*}(g_5) \otimes \pi_{\phi}(g_6) \otimes \pi_{\phi^*}(g_6) \end{aligned} \quad (109)$$

$$\begin{aligned} \sigma_4(g_1, g_2, g_3, g_4) &= \pi_{\alpha}(g_1) \otimes \pi_{\alpha^*}(g_4) \otimes \pi_{\beta}(g_1) \otimes \pi_{\beta^*}(g_2) \otimes \pi_{\gamma}(g_3) \otimes \pi_{\gamma^*}(g_1) \otimes \pi_{\delta}(g_3) \otimes \pi_{\delta^*}(g_2) \otimes \pi_{\epsilon}(g_4) \otimes \pi_{\epsilon^*}(g_3) \otimes \pi_{\phi}(g_2) \otimes \pi_{\phi^*}(g_4) \\ &\quad \cdots \otimes \pi_{\delta}(g_3) \otimes \pi_{\delta^*}(g_2) \otimes \pi_{\epsilon}(g_4) \otimes \pi_{\epsilon^*}(g_3) \otimes \pi_{\phi}(g_2) \otimes \pi_{\phi^*}(g_4) \end{aligned} \quad (110)$$

$$\begin{aligned} \sigma_1(g) &= \pi_{\alpha}(g) \otimes \pi_{\alpha^*}(g) \otimes \pi_{\beta}(g) \otimes \pi_{\beta^*}(g) \otimes \pi_{\gamma}(g) \otimes \pi_{\gamma^*}(g) \otimes \pi_{\delta}(g) \otimes \pi_{\delta^*}(g) \otimes \pi_{\epsilon}(g) \otimes \pi_{\epsilon^*}(g) \otimes \pi_{\phi}(g) \otimes \pi_{\phi^*}(g) \\ &\quad \cdots \otimes \pi_{\delta}(g) \otimes \pi_{\delta^*}(g) \otimes \pi_{\epsilon}(g) \otimes \pi_{\epsilon^*}(g) \otimes \pi_{\phi}(g) \otimes \pi_{\phi^*}(g) \end{aligned} \quad (111)$$

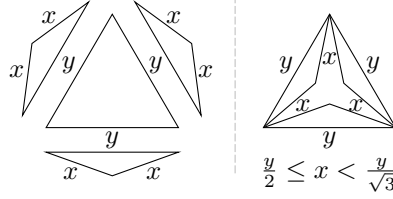


Figure 5: A example of a sextuple of edge lengths, $(a, b, c, d, e, f) = (y, y, y, x, x, x)$, which satisfies all triangle inequality closure constraints when $y \leq 2x$, but cannot be the edge lengths of any Euclidean tetrahron when $\sqrt{3}x < y$.

At this stage, it is worth noticing that σ_1 is the representation of $\text{GL}(d)$ which is obtained by restricting σ_4 (respectively σ_6) to the diagonal subgroup $\text{GL}(d) \subseteq \text{GL}(d)^{\times 4}$ (respectively $\text{GL}(d) \subseteq \text{GL}(d)^{\times 6}$). Now consider the subspace of $\mathcal{L}$ consisting of vectors invariant under the action of $\text{GL}(d)^{\times 4}$ (via σ_4), i.e.,

$$\mathcal{V}_4 := (\mathcal{L})^{\text{GL}(d)^{\times 4}} = \{w \in \mathcal{L} \mid \forall g_1, g_2, g_3, g_4 \in \text{GL}(d) : \sigma_4(g_1, g_2, g_3, g_4)w = w\}. \quad (112)$$

Whenever it happens to be that $\mathcal{V}_4 \neq 0$, then one may conclude

$$\begin{aligned} (V_\alpha \otimes V_\gamma^* \otimes V_\beta)^{\text{GL}(d)} &\neq 0, \\ (V_\beta^* \otimes V_\phi \otimes V_\delta^*)^{\text{GL}(d)} &\neq 0, \\ (V_\delta \otimes V_\gamma \otimes V_{\epsilon^*})^{\text{GL}(d)} &\neq 0, \\ (V_\epsilon \otimes V_\phi^* \otimes V_\alpha^*)^{\text{GL}(d)} &\neq 0, \end{aligned} \quad \text{\texttt{\{eq:face_inv\}}} \quad (113)$$

which, by ??, implies the existence of Hermitian operators in $\mathcal{M}$

$$A, A' \in \mathcal{O}_\alpha, \quad B, B' \in \mathcal{O}_\beta, \quad C, C' \in \mathcal{O}_\gamma, \quad D, D' \in \mathcal{O}_\delta, \quad E, E' \in \mathcal{O}_\epsilon, \quad F, F' \in \mathcal{O}_\phi, \quad \text{\texttt{\{eq:prime_unprime\}}} \quad (114)$$

satisfying the following four conditions

$$A - C' + B = 0, \quad -B' + F - D' = 0, \quad D + C - E' = 0, \quad E - F' - A' = 0. \quad (115)$$

From this perspective, it becomes clear that the existence of non-zero $\text{GL}(d)^{\times 4}$ -invariant vectors in $\mathcal{L}$ is a necessary condition for $\alpha, \beta, \gamma, \delta, \epsilon$, and ϕ to be tetrahedral spectra. Unfortunately, it can be shown that this condition is indeed insufficient; the insufficiency can be attributed to the fact that Eq. (113) does not require that the primed and unprimed Hermitian matrices are in Eq. (114) are the same, i.e., that $A = A'$, $B = B'$, etc.

Remark 8. To address the insufficiency of Eq. (113), consider the subspace of $\mathcal{L}$ consisting of vectors invariant under the action of $\text{GL}(d)^{\times 6}$ (via σ_6), i.e.,

$$\mathcal{V}_6 := (\mathcal{L})^{\text{GL}(d)^{\times 6}} = \{w \in \mathcal{L} \mid \forall g_1, g_2, g_3, g_4, g_5, g_6 \in \text{GL}(d) : \sigma_6(g_1, g_2, g_3, g_4, g_5, g_6)w = w\}. \quad (116)$$

For all dominant weights λ , Schur's lemma implies

$$(V_\lambda \otimes V_\lambda^*)^{\text{GL}(d)} \simeq \text{Hom}_{\text{GL}(d)}(V_\lambda, V_\lambda) \simeq \mathbb{C} \text{id}_{V_\lambda}, \quad (117)$$

which in turn implies the subspace of $\text{GL}(d)^{\times 6}$ -invariant vectors in $\mathcal{L}$ is always one-dimensional, $\dim(\mathcal{V}_6) = 1$. In any case, if there exists a vector which is *both* $\text{GL}(d)^{\times 4}$ and $\text{GL}(d)^{\times 6}$, i.e., $w \in \mathcal{V}_6 \cap \mathcal{V}_4 \neq 0$, then it can be shown (by similar arguments to the proof of ??⁶) that there exists Hermitian operators

$$A \in \mathcal{O}_\alpha, \quad B \in \mathcal{O}_\beta, \quad C \in \mathcal{O}_\gamma, \quad D \in \mathcal{O}_\delta, \quad E \in \mathcal{O}_\epsilon, \quad F \in \mathcal{O}_\phi, \quad (118)$$

⁶Specifically, one begins with a generic twelve-fold coherent product vector $v_0 \in \mathcal{L}$ and iteratively flowing along gradients to minimize the norm of $\sigma_6(g'_i)\sigma_4(g_j)v_{n-1} \in \mathcal{L}$ with respect to g'_i and g_j to obtain $v_n \in \mathcal{L}$. The resulting sequence of points exhibits a limit point $\tilde{v}$ which is (i) non-zero (because $\langle w, \tilde{v} \rangle = \langle w, v_n \rangle = \langle w, v_0 \rangle \neq 0$), and (ii) minimal in norm with respect to both orbits, i.e. both moment maps vanish: $\mu_4([\tilde{v}]) = 0$ and $\mu_6([\tilde{v}]) = 0$. Thanks to Harold for sketching out this argument for me.

satisfying the following four conditions

$$A - C + B = 0, \quad -B + F - D = 0, \quad D + C - E = 0, \quad E - F - A = 0, \quad (119)$$

which are readily seen to be equivalent to tetrahedral constraints in Eq. (3). In other words, the existence of a $\mathrm{GL}(d)^{\times 4}$ and $\mathrm{GL}(d)^{\times 6}$ invariant vector in $\mathcal{L}$ is sufficient to conclude that the sextuple $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral spectra. However, using the results of [Rob99], it can be concluded for most tetrahedral $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ that no “doubly-invariant” vectors exist in $\mathcal{L}$, i.e. $\mathcal{V}_4 \cap \mathcal{V}_6 = 0$.

Remark 9. In [Rob99], Roberts originally used the stationary phase method to prove the Ponzano and Regge semiclassical formula [PR68] for the $6j$ coefficients (i.e., the $d = 2$ recoupling symbol). In order to apply the technique of stationary phases, it must be the case that exponent in the integrand has *isolated* critical points. For the case of the $6j$ symbol (with generic values for the six j s), there were two isolated critical points corresponding to (i) the $\mathrm{SO}(3)$ -orbit $[\tau]$ of a Euclidean tetrahedron τ with fixed side lengths and (ii) the $\mathrm{SO}(3)$ -orbit $[\tau']$ of the Euclidean tetrahedron τ' which is the mirror-image of the tetrahedron τ with the same fixed side lengths.

TODO: For the more general case of $d \geq 3$, it is unclear to me if the analogous critical points (whose union is identified with all Hermitian matrix-valued solutions to the tetrahedral equations) correspond to a finite number of disjoint coadjoint orbits under the subgroup $\mathrm{GL}(d)$ (as in the $d = 2$ case). As a result, at the moment, it is unclear to me how to establish an asymptotic formula $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ in the “scaling $\lambda \mapsto k\lambda$ ” regime. Instead, it seems the next best thing is to use the “completeness-type” relations between various recoupling symbols to prove the existence of a convergent sequence of weights such that the corresponding sequence of recoupling symbols admits of a power-law type lower bound.

Even if we were to stipulate/conjecture/assume that $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are such that $(\mu_E^{-1}(0) \cap \mu_V^{-1}(0)) / \mathrm{GL}(d)$ consists of a finite number of isolated fixed points (as in the $d = 2$, non-degenerate case considered by Roberts, it would be a great deal of work to compute the Hessians and phases at each of the critical points in order to formally check that the phases are *NOT* “misaligned” in the sense that the complex contributions destructively interfere leading to a trivial lower-bound on the norm of $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ of zero.

Even worse, even in the $d = 2$ case, one generically expects non-isolated critical points even after integrating over a global $\mathrm{GL}(d)$ because of the existence of non-rigid polyhedra with fixed edge lengths (the tetrahedron is locally rigid, while a cube is *not* locally rigid in this sense).

The issues which arise when attempting to generalize the scaling $\lambda \mapsto k\lambda$ asymptotics are very similar to the issues which arise when attempting to prove Mathieu’s conjecture for K -finite functions $f(u) = \langle w, \pi(u)v \rangle$ in the non-positive-definite case, i.e., $w \neq v$ [FW20].

3.7 Schur-Weyl duality & decomposition of tensor powers

`eq:schur_weyl_and_tensor_powers`

The purpose of this section is to derive some upper and lower bounds on generalized $6j$ symbol for $\mathrm{SU}(d)$ in relation to the existence or non-existence of Hermitian operators satisfying the tetrahedral relations Eq. (3). This section essentially constitutes a generalization of the proof techniques in the “quantum-information-theoretic” approach taken in [Chr08]. Specifically, the results of this section are derived from considering a type of pseudo-commutative binomial expansion of tensor powers of sums of Hermitian operators and applying spectrum estimation to the resulting terms.

Throughout this section, we make use of the following notation.

Definition 5 (Schur-Weyl decomposition). Let n and d be positive integers and

$$(\mathbb{C}^d)^{\otimes n} \cong \bigoplus_{\lambda \vdash_d n} W_\lambda \otimes V_\lambda^d \quad (\text{eq:schur_weyl_decomp}) \quad (120)$$

be the Schur-Weyl decomposition of $(\mathbb{C}^d)^{\otimes n}$ into irreducible representations of $S_n \times \mathrm{SU}(d)$. Moreover, let P_λ denote the Hermitian projection operator which projects onto the isotypic subspace indexed by $\lambda \vdash_d n$ in Eq. (120)

The following lemma places an upper bound on the value of $\mathrm{Tr}(P_\lambda X^{\otimes n})$ for given λ and positive semidefinite Hermitian matrix X [Chr08; KW01].

Proposition 16 (Spectrum estimation). *Let $\lambda \vdash_d n$ and let X be a positive semidefinite matrix with eigenvalues χ . Then*

$$\mathrm{Tr}(P_\lambda X^{\otimes n}) \leq \dim(V_\lambda^d) \exp(-nD(\bar{\lambda} \parallel \bar{\chi}) + n \log |\chi|), \quad (121)$$

where $D(\bar{\lambda} \parallel \bar{\chi})$ denotes the Kullbeck-Leibler divergence of $\bar{\lambda}$ from $\bar{\chi}$.

Proof. See [KW01] for a proof. Also note that the $n \log |\chi|$ term simply accounts for the possibility that $\mathrm{Tr}(X) \neq 1$ because $\mathrm{Tr}(X)^n = |\chi|^n = \exp(n \log |\chi|)$. $\square$

Following [Chr08], we will be interesting in applying the spectrum estimation result (Proposition 16) to Hermitian operators of the form $C = A + B$.

lem:binomial_expansion

Lemma 17. *Let n be a positive integer and let A, B, C be $d \times d$ Hermitian operators satisfying $A + B = C$. Then for all $0 \leq k \leq n$,*

$$\mathrm{Tr}(P_\lambda(A^{\otimes k} \otimes B^{\otimes n-k})) \leq \binom{n}{k}^{-1} \mathrm{Tr}(P_\lambda C^{\otimes n}). \quad (122)$$

Proof. The proof follows from the (non-commutative) binomial expansion of $C^{\otimes n} = (A + B)^{\otimes n}$ together with the S_n -equivariance of P_λ , which yields

$$\mathrm{Tr}(P_\lambda C^{\otimes n}) = \sum_{k=0}^n \binom{n}{k} \mathrm{Tr}(P_\lambda(A^{\otimes k} \otimes B^{\otimes n-k})). \quad (123)$$

Since each of the summands above are non-negative, for all $0 \leq k \leq n$, we conclude

$$\mathrm{Tr}(P_\lambda C^{\otimes n}) \geq \binom{n}{k} \mathrm{Tr}(P_\lambda(A^{\otimes k} \otimes B^{\otimes n-k})), \quad (124)$$

which is equivalent to the claimed inequality. $\square$

Repeatedly combining Lemma 17 with Proposition 16 produces the following result.

Theorem 18. *Let (A, B, C, D, E, F) be a tetrahedral sextuple of positive semidefinite $d \times d$ Hermitian matrices with eigenvalues $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$. Fix x, y, z positive integers and define*

$$\Omega := \Omega_{A,B,D}^{x,y,z} = A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z}. \quad (125)$$

Moreover, let $(\alpha', \beta', \gamma', \delta', \epsilon', \phi')$ be Young diagrams of the following sizes:

$$\begin{aligned} \alpha' \vdash_d x, & \quad \beta' \vdash_d y, & \quad \gamma' \vdash_d x + y, \\ \delta' \vdash_d z, & \quad \epsilon' \vdash_d x + y + z, & \quad \phi' \vdash_d y + z, \end{aligned} \quad (126)$$

and define corresponding projection operators acting on $(\mathbb{C}^d)^{\otimes x+y+z}$ by

$$\begin{aligned} \Pi_{\alpha'} &:= P_{\alpha'} \otimes \mathrm{id}^{\otimes y} \otimes \mathrm{id}^{\otimes z}, & \Pi_{\beta'} &:= \mathrm{id}^{\otimes x} \otimes P_{\beta'} \otimes \mathrm{id}^{\otimes z}, & \Pi_{\gamma'} &:= P_{\gamma'} \otimes \mathrm{id}^{\otimes x} \otimes \mathrm{id}^{\otimes z}, \\ \Pi_{\delta'} &:= \mathrm{id}^{\otimes x} \otimes \mathrm{id}^{\otimes y} \otimes P_{\delta'}, & \Pi_{\epsilon'} &:= P_{\epsilon'}, & \Pi_{\phi'} &:= \mathrm{id}^{\otimes x} \otimes P_{\phi'}. \end{aligned} \quad (127)$$

TODO: Figure out the rest of the theorem and proof (i.e. the upper bounding rate function). In particular, what should x, y, z be as a function of n ? How to approximate the “trace distribution”. The essential idea is to take a large positive integer n and let x, y, z satisfy $x \approx n|\alpha|/|\epsilon|$, $y \approx n|\beta|/|\epsilon|$, and $z \approx n|\delta|/|\epsilon|$. By choosing the values of x, y, z appropriately, the hope is that one can lower-bound the multinomial distribution terms appearing in the upper bounds in the proof below by polynomial-like terms in n .

Proof. Consider the Hilbert-Schmidt inner product,

$$\langle \Pi_{\lambda'}, \bar{\Omega} \rangle = \text{Tr}(\Pi_{\lambda'} \bar{\Omega}), \quad (128)$$

between the normalisation of Ω ($\bar{\Omega} := \Omega / \text{Tr}(\Omega)$) and each of the projection operators $\Pi_{\lambda'}$ defined by Eq. (127). It is possible to upper-bound $\langle \Pi_{\lambda'}, \bar{\Omega} \rangle$ using a combination of the upper bound coming from Lemma 17 and the upper bound coming from the spectrum estimation technique Proposition 16. For a detailed example, take $\text{Tr}(\Pi_{\gamma'} \bar{\Omega})$ while noting that $C = A + B$ and thus $|\gamma| = |\alpha| + |\beta|$:

$$\text{Tr}(\Pi_{\gamma'} \bar{\Omega}) = \frac{\text{Tr}[(P_{\gamma'} \otimes \text{id}^{\otimes z})(A^{\otimes x} \otimes B^{\otimes y} \otimes D^{\otimes z})]}{|\alpha|^x |\beta|^y |\delta|^z}, \quad (129)$$

$$= \frac{\text{Tr}[P_{\gamma'}(A^{\otimes x} \otimes B^{\otimes y})] |\delta|^z}{|\alpha|^x |\beta|^y |\delta|^z}, \quad (130)$$

$$\leq \frac{1}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \text{Tr}[P_{\gamma'} C^{\otimes x+y}], \quad (\text{Lemma 17}) \quad (131)$$

$$\leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)D(\bar{\gamma}' \| \bar{\gamma})), \quad (\text{Proposition 16}). \quad (132)$$

In summary, we obtain the following set of upper bounds :

$$\text{Tr}(\Pi_{\alpha'} \bar{\Omega}) \leq \dim(V_{\alpha'}^d) \exp(-xD(\bar{\alpha}' \| \bar{\alpha})), \quad (133)$$

$$\text{Tr}(\Pi_{\beta'} \bar{\Omega}) \leq \dim(V_{\beta'}^d) \exp(-yD(\bar{\beta}' \| \bar{\beta})), \quad (134)$$

$$\text{Tr}(\Pi_{\gamma'} \bar{\Omega}) \leq \frac{(|\alpha| + |\beta|)^{x+y}}{|\alpha|^x |\beta|^y} \frac{x!y!}{(x+y)!} \dim(V_{\gamma'}^d) \exp(-(x+y)D(\bar{\gamma}' \| \bar{\gamma})), \quad (135)$$

$$\text{Tr}(\Pi_{\delta'} \bar{\Omega}) \leq \dim(V_{\delta'}^d) \exp(-zD(\bar{\delta}' \| \bar{\delta})), \quad (136)$$

$$\text{Tr}(\Pi_{\epsilon'} \bar{\Omega}) \leq \frac{(|\alpha| + |\beta| + |\delta|)^{x+y+z}}{|\alpha|^x |\beta|^y |\delta|^z} \frac{x!y!z!}{(x+y+z)!} \dim(V_{\epsilon'}^d) \exp(-(x+y+z)D(\bar{\epsilon}' \| \bar{\epsilon})), \quad (137)$$

$$\text{Tr}(\Pi_{\phi'} \bar{\Omega}) \leq \frac{(|\beta| + |\delta|)^{y+z}}{|\beta|^y |\delta|^z} \frac{y!z!}{(y+z)!} \dim(V_{\phi'}^d) \exp(-(y+z)D(\bar{\phi}' \| \bar{\phi})). \quad (138)$$

TODO: see the **TODO** in the theorem.

thm:seq_ud_recoupling_norm_lower_bound $\square$

Theorem 19. Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ be the eigenvalues of positive semidefinite Hermitian matrices (A, B, C, D, E, F) satisfying Eq. (3). Then there exists a sextuple of sequences of Young diagrams $(\alpha'(n), \beta'(n), \gamma'(n), \delta'(n), \epsilon'(n), \phi'(n))$ satisfying

$$\begin{aligned} \lim_{n \rightarrow \infty} \alpha'(n) &= \bar{\alpha}, & \lim_{n \rightarrow \infty} \beta'(n) &= \bar{\beta}, & \lim_{n \rightarrow \infty} \gamma'(n) &= \bar{\gamma}, & \text{\textit{seq:sextuple_convergence}} \\ \lim_{n \rightarrow \infty} \delta'(n) &= \bar{\delta}, & \lim_{n \rightarrow \infty} \epsilon'(n) &= \bar{\epsilon}, & \lim_{n \rightarrow \infty} \phi'(n) &= \bar{\phi}, & (139) \end{aligned}$$

$$\lim_{n \rightarrow \infty} \frac{|\alpha(n)|}{|\epsilon(n)|} = \frac{|\alpha|}{|\epsilon|}, \quad \lim_{n \rightarrow \infty} \frac{|\beta(n)|}{|\epsilon(n)|} = \frac{|\beta|}{|\epsilon|}, \quad \lim_{n \rightarrow \infty} \frac{|\delta(n)|}{|\epsilon(n)|} = \frac{|\delta|}{|\epsilon|}, \quad (140)$$

and

$$\left\| \begin{Bmatrix} \alpha(n) & \beta(n) & \gamma(n) \\ \delta(n) & \epsilon(n) & \phi(n) \end{Bmatrix} \right\|_{\text{HS}}^2 \geq \frac{1}{\text{poly}(?n?)} \quad \text{\textit{seq:seq_ud_recoupling_norm_lower_bound}} \quad (141)$$

TODO: Figure out the precise statement of the lower bound; it depends on how rapidly x, y, z jump with respect to n .

3.8 The asymptotics of irrep dimensions

It will be useful for what follows to better understand how the dimension, $\dim(V_{\lambda}^d)$, of the irreducible representation V_{λ}^d of $\text{SL}(d)$ grows as d becomes much larger $|\lambda|$ while $|\lambda|$ itself is much larger than $\ell(\lambda)$. Itzykson and Nauenberg provide the following estimate for $\dim(V_{\lambda}^d)$ [IN66].

lem:in_estimate

Lemma 20. *Let λ be an positive integer partition of length $\ell := \ell(\lambda)$, let $d \geq \ell$ be a positive integer and let $\rho^d := (d-1, d-2, \dots, 2, 1)$ be the Weyl-vector for $\text{SL}(d)$. The following approximation holds when $d \gg |\lambda|^2$ and $\lambda_\ell \gg \ell$:*

$$\dim(V_\lambda^d) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times d^{|\lambda|} \times \left(1 + O\left(\frac{|\lambda|^2}{d}\right)\right). \quad (142) \quad \text{\texttt{eq:in_estimate}}$$

Proof. Weyl's character formula can be used to derive the following formula for $\dim(V_\lambda^d)$ in terms of the Young diagram λ and the dimension d [IN66]:

$$\dim(V_\lambda^d) = \frac{\Delta(\lambda + \rho^d)}{\Delta(\rho^d)} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i + \rho_i^d) - (\lambda_j + \rho_j^d)}{\rho_i^d - \rho_j^d} = \prod_{1 \leq i < j \leq d} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i}, \quad (143)$$

where $\Delta(x) = \prod_{i < j} (x_i - x_j)$ is the Vandermonde determinant. Isolating the terms above which depend on d from those that do not yields

$$\dim(V_\lambda^d) = \dim(V_\lambda^\ell) \times \prod_{i=1}^{\ell} \frac{\rho_i^\ell!}{(\lambda_i + \rho_i^\ell)!} \times \prod_{i=1}^{\ell} \frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!}. \quad (144) \quad \text{\texttt{eq:dim_separation_large_d}}$$

Given positive integers a and b such that $b \gg a^2$, one obtains the following approximation:

$$\begin{aligned} \log \left(\frac{(a+b)!}{b!} \right) &= \log \prod_{j=1}^a (b+j) = \sum_{j=1}^a \log(b+j) \\ &= a \log b + \sum_{j=1}^a \log \left(1 + \frac{j}{b} \right) = a \log b + O\left(\frac{a^2}{b}\right). \end{aligned} \quad (145)$$

The assumption that $d \gg |\lambda|^2$ and $i \leq \ell \ll \lambda_\ell \leq |\lambda|$ also implies $d - i = \rho_i^d \gg \lambda_i^2$. Therefore,

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log \rho_i^d + O\left(\frac{\lambda_i^2}{\rho_i^d}\right). \quad (146)$$

The assumptions also imply the following simplification

$$\log \left(\frac{(\lambda_i + \rho_i^d)!}{\rho_i^d!} \right) = \lambda_i \log d + O\left(\frac{|\lambda|^2}{d}\right) \quad (147)$$

which, when applied to Eq. (144), proves the claim. $\square$

Using Lemma 20, it can be seen that for fixed λ , $\dim(V_\lambda^d)$ generally grows polynomially in d with leading exponent $|\lambda|$ and leading coefficient depending only on λ . The objective of the following lemma is to characterize the asymptotics of this leading coefficient for large $|\lambda|$ and even large d . **TODO: Figure out a elegant way to present these “double asymptotics”, maybe just epsilonifying everything, like $\max\{\ell/|\lambda|, |\lambda|/d\} \leq \epsilon$.** *lem:double_asymptotics*

Lemma 21. *Let everything be as in Lemma 20. Then*

$$\log \dim(V_\lambda^d) = |\lambda| + |\lambda| \log(d) + s(\lambda) + O(\ell^2 \log |\lambda|) + O\left(\frac{|\lambda|^2}{d}\right) \quad (148)$$

where $s(\lambda)$ is defined by

$$s(\lambda) := - \sum_{i=1}^{\ell(\lambda)} \lambda_i \log \lambda_i. \quad (149) \quad \text{\texttt{eq:unnorm_shannon}}$$

Proof. The proof proceeds by considering the multiplicands in Eq. (142) separately. First, we upper bound the dimension $\dim(V_\lambda^\ell)$; for indices $i < j \leq \ell$, one has $(\lambda_i - i) - (\lambda_j - j) \leq |\lambda| + 1$ as well as $j - i \geq 1$ which implies

$$\dim(V_\lambda^\ell) = \prod_{1 \leq i < j \leq \ell} \frac{(\lambda_i - i) - (\lambda_j - j)}{j - i} \leq (|\lambda| + 1)^{\ell(\ell-1)/2}, \quad \text{\textcolor{blue}{eq:dim_lambda_ell_bound}} \quad (150)$$

and thus $\log \dim(V_\lambda^\ell) = O(\ell^2 \log |\lambda|)$.

Second, we apply Stirling's approximation [Rob55] to the second term in Eq. (142) under the given assumption that $\lambda_\ell \gg \ell$ (which implies $\rho_i^\ell \ll \lambda_i$),

$$\log \left(\prod_{i=1}^{\ell} \frac{\rho_i^{\ell!}}{(\lambda_i + \rho_i^\ell)!} \right) = \sum_{i=1}^{\ell} \log \rho_i^{\ell!} - \sum_{i=1}^{\ell} \log(\lambda_i + \rho_i^\ell)!, \quad (151)$$

$$= |\lambda| + \sum_{i=1}^{\ell} (\rho_i^\ell \log \rho_i^\ell - (\lambda_i + \rho_i^\ell) \log(\lambda_i + \rho_i^\ell) + O(\log \lambda_i)), \quad (152)$$

$$= |\lambda| - \sum_{i=1}^{\ell} \lambda_i \log \lambda_i + O(\ell \log |\lambda|), \quad (153)$$

$$\text{\textcolor{blue}{eq:unnorm_shannon_term}} \quad (154)$$

Combining Eq. (150) and Eq. (154) with Lemma 20 proves the claim. $\square$

Remark 10. Note that the quantity $s(\lambda)$ defined in Eq. (149) resembles the Shannon entropy,

$$S(p) = - \sum_{i=1}^{\ell} p_i \log p_i \quad (155)$$

of a probability distribution $p = (p_1, \dots, p_\ell)$. Indeed, $s(\lambda)$ is related to the Shannon entropy of the *normalization* of λ by the formula

$$s(\lambda) = |\lambda| S \left(\frac{\lambda}{|\lambda|} \right) - |\lambda| \log |\lambda|. \quad \text{\textcolor{blue}{eq:little_s_relation}} \quad (156)$$

TODO: Perhaps a more direct way of relating the asymptotics of $\dim(V_\lambda^d)$ to $\dim(W_\lambda)$ can be found in Harrow's paper [Har23, Lemma 2]:

Lemma 22. *Let λ be a Young diagram with at most d parts. Then*

$$\frac{|\lambda|!}{d^{|\lambda|}} \frac{\dim(V_\lambda^d)}{\dim(W_\lambda)} = \prod_{(i,j) \in \lambda} \left(1 + \frac{j-i}{d} \right) \quad (157)$$

3.9 An entropic inequality for tetrahedral spectra

thm:ud_recouple_bound

Theorem 23. *Let $(\alpha, \beta, \gamma, \delta, \epsilon, \phi) \in \mathbb{Z}_{\text{dec}}^d$ be a sextuple of dominant weights for $\text{GL}(d)$ and let $\left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\}$ be the corresponding $\text{GL}(d)$ -recoupling operator. Then*

$$\left\| \left\{ \begin{smallmatrix} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{smallmatrix} \right\} \right\|_{HS}^2 \leq \left(c_{\alpha, \phi}^\epsilon c_{\beta, \delta}^\phi c_{\alpha, \beta}^\gamma c_{\gamma, \delta}^\epsilon \right)^{\frac{1}{2}} \frac{\dim(V_\gamma) \dim(V_\phi)}{\dim(V_\beta) \dim(V_\epsilon)} \quad (158)$$

Proof. TODO: Explain what this all means carefully. Applying both the Cauchy-Schwarz inequality with

respect to the contraction annotated below, and Schur's lemma multiple times yields the following inequality:

$$\begin{aligned}
& \textcircled{\epsilon}^2 C_{\alpha,\beta}^\gamma C_{\alpha,\phi}^\epsilon C_{\beta,\delta}^\phi C_{\gamma,\delta}^\epsilon = \\
& \left[\text{Diagram 1} \right] = \left[\text{Diagram 2} \right] \\
& \leq \left[\text{Diagram 3} \right]^{\frac{1}{2}} = \frac{1}{\textcircled{\beta} \textcircled{\epsilon}} \left[\text{Diagram 4} \right]^{\frac{1}{2}} \\
& = \frac{\textcircled{\epsilon}^2 \textcircled{\gamma} \textcircled{\phi}}{\textcircled{\beta} \textcircled{\epsilon}} \left[\textcircled{C_{\alpha,\phi}^\epsilon} \textcircled{C_{\beta,\delta}^\phi} \textcircled{C_{\alpha,\beta}^\gamma} \textcircled{C_{\gamma,\delta}^\epsilon} \right]^{\frac{1}{2}}
\end{aligned} \tag{159}$$

lem:asymptotic_dimension_ratio $\square$

Lemma 24. Let $\gamma, \phi, \beta, \epsilon$ be a quadruple of Young diagrams such that

$$|\gamma| + |\phi| = |\beta| + |\epsilon| \tag{160}$$

{eq:balanced_sizes}

and that $|\epsilon| = \max\{|\gamma|, |\phi|, |\beta|, |\epsilon|\}$. Let $\ell := \max\{\ell(\gamma), \ell(\phi), \ell(\beta), \ell(\epsilon)\}$ and let d be a positive integer such that $d \geq \ell$. Then

$$\log \left(\frac{\dim(V_\gamma^d) \dim(V_\phi^d)}{\dim(V_\beta^d) \dim(V_\epsilon^d)} \right) = I(\gamma, \phi; \beta, \epsilon) + O(\ell^2 \log |\epsilon|) + O \left(\frac{\textcircled{\epsilon}^2 \textcircled{\gamma} \textcircled{\phi}}{d} \right) \tag{161}$$

{eq:asymptotic_dimension_ratio}

where $I(\gamma, \phi; \beta, \epsilon)$ is the function

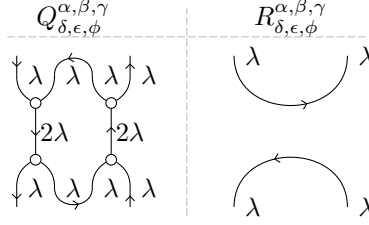
$$I(\gamma, \phi; \beta, \epsilon) := s(\gamma) + s(\phi) - s(\beta) - s(\epsilon). \tag{162}$$

Moreover, $I(k\gamma, k\phi; k\beta, k\epsilon) = kI(\gamma, \phi; \beta, \epsilon)$ for any $k > 0$.

Proof. The proof follows from four applications of Lemma 21. The assumption that the sizes of the Young diagrams are balanced in the sense of Eq. (160) implies that the $\log(d)$ terms appearing in Lemma 21 cancel each other and one obtains Eq. (161).

Using the strong-duality theorem and central-limit theorem results [FW20, Theorem 1 & Proposition 3.1], one can prove that if $(\alpha, \beta, \gamma, \delta, \epsilon, \phi)$ are tetrahedral, then $Q_{k\delta, k\epsilon, k\phi}^{k\alpha, k\beta, k\gamma}$ exhibits a power-law in k lower bound, otherwise they are not tetrahedral and then $Q_{k\delta, k\epsilon, k\phi}^{k\alpha, k\beta, k\gamma}$ exhibits exponential decay in k .

The primary difference between $Q_{\delta, \epsilon, \phi}^{\alpha, \beta, \gamma}$ and $R_{\delta, \epsilon, \phi}^{\alpha, \beta, \gamma}$ is diagrammatically expressed in the following figure:



TC: Developing a better understand of the relationship between these two operators would be beneficial for proving cleaner and tighter results about the recoupling operator.

4 The spectral quantum marginal problem

The purpose of this section will be to re-articulate the spectral embedding defined in [CSW18] and state the sense in which the tetrahedral spectra problem is an instance of the quantum marginal problem. From here we can discuss the extent to which known inequalities coming from quantum information theory yield constraints necessarily satisfied by tetrahedral spectra.

4.1 A spectral embedding

defn:spec_embed

Definition 6. Let $(H_1, \dots, H_k) \in \mathcal{H}(d)^{\times k}$ be a k -tuple of $d \times d$ positive-semidefinite Hermitian matrices and let X be a subset of $\{1, \dots, k\}$. Let $\{|1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^d$ and $\{|0\rangle, |1\rangle, \dots, |d\rangle\}$ be an orthonormal basis for $\mathbb{C}^{d+1}$. Define $H_X := \sum_{i \in X} H_i$ as the partial sum. Define a Hermitian operator, Φ_X , acting on $(\mathbb{C}^{d+1})^{\otimes k}$ with coefficients with respect to the computational bases $|i_1 \dots i_k\rangle$ be given by the following formula:⁷

$$\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle = \begin{cases} \langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle & \exists! i_p, j_q > 0, p, q \in X \\ 0 & \text{otherwise.} \end{cases} \quad (169)$$

Note that, up to shuffling the $k+1$ copies of $\mathbb{C}^{d+1}$ in $(\mathbb{C}^{d+1})^{\otimes k}$, Φ_X is of the form $\tilde{\Phi}_X \otimes Z_{[k] \setminus X}$ where

$$Z_X = |0 \dots 0\rangle \langle 0 \dots 0|. \quad (170)$$

and where $\tilde{\Phi}_X$ acts on $(\mathbb{C}^{d+1})^{\otimes |X|}$. Also define the operator Γ_X which is $\tilde{\Phi}_X$ shifted by Z_X

$$\Gamma_X := \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \quad (171)$$

and in particular $\Gamma_\emptyset := 1$.

Example 6. Given two Hermitian operators $(H_1, H_2) \in \mathcal{H}(d)^{\times 2}$, the matrix $\Gamma_{1,2}$ has the following coefficients for all $i, j, k, l \in \{1, \dots, d\}$.

$$\langle 00 | \Gamma_{1,2} | 00 \rangle = 1 - \text{Tr}(H_{1,2}), \quad (172)$$

$$\langle i0 | \Gamma_{1,2} | j0 \rangle = \langle i | H_1 | j \rangle, \quad (173)$$

$$\langle 0i | \Gamma_{1,2} | 0j \rangle = \langle i | H_2 | j \rangle, \quad (174)$$

$$\langle i0 | \Gamma_{1,2} | 0j \rangle = \langle i | H_1^{\frac{1}{2}} H_2^{\frac{1}{2}} | j \rangle, \quad (175)$$

$$\langle 0i | \Gamma_{1,2} | j0 \rangle = \langle i | H_2^{\frac{1}{2}} H_1^{\frac{1}{2}} | j \rangle, \quad (176)$$

$$\langle ik | \Gamma_{1,2} | jl \rangle = 0. \quad (177)$$

lem:spec_embed

Lemma 26. Let everything be defined as in Definition 6. The follow statements hold for all subsets $X \subseteq [k]$.

(i) Up to padding by zeros, the eigenvalues of $\tilde{\Phi}_X$, Φ_X and H_X are all identical.

$$\lambda(\tilde{\Phi}_X) \simeq \lambda(\Phi_X) \simeq \lambda(H_X). \quad (178)$$

(ii) The eigenvalues of Γ_X are equal to the eigenvalues of $\tilde{\Phi}_X$ plus the additional value of $1 - \text{Tr}(H_X)$.

$$\lambda(\Gamma_X) \simeq \lambda(\tilde{\Phi}_X) \oplus \{1 - \text{Tr}(H_X)\}. \quad (179)$$

(iii) The partial trace of $\Gamma_{[k]}$ over $[k] \setminus X$ is equal to Γ_X :

$$\text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) = \Gamma_X. \quad (180)$$

⁷Specifically, $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ is non-zero only if there exists a unique $p \in [k]$ such that $i_p > 0$ and a unique $q \in [k]$ such that $j_q > 0$ (with all other indices $i_{p'}$ and $j_{q'}$ for $p' \neq p$ and $q' \neq q$ being zero). Furthermore, if $p, q \in [k]$ are the unique such indices, then $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ is zero if either $p \notin X$ or $q \notin X$; otherwise $p, q \in X$ and then $\langle i_1 \dots i_k | \Phi_X | j_1 \dots j_k \rangle$ takes the value $\langle i_p | H_p^{\frac{1}{2}} H_q^{\frac{1}{2}} | j_q \rangle$, the inner product between the i_p th column of $H_p^{\frac{1}{2}}$ and the j_q th column of $H_q^{\frac{1}{2}}$.

Proof. (i) From the definition of Φ_X , one observes that $\langle i_1 \cdots i_k | \Phi_X | i_1 \cdots i_k \rangle$ is only non-zero if there exists a unique non-zero $i_p \neq 0$ where $p \in X$ and thus

$$\text{Tr}(\Phi_X) = \sum_{p \in X} \sum_{j=1}^d \langle j | H_p | j \rangle = \sum_{p \in X} \text{Tr}(H_p) = \text{Tr}(H_X). \quad (181)$$

More generally, for any positive integer $\ell \in \mathbb{N}_{>0}$, one observes that $\text{Tr}(\Phi_X^\ell) = \text{Tr}(H_X^\ell)$. (**TODO: Prove this directly.**) As the eigenvalues of any matrix X are determined (up to padding by zeros) by the values of $\text{Tr}(X^\ell)$, we conclude that Φ_X and H_X have the same eigenvalues (up to padding by zeros) (**TODO: Is there a more direct way to prove this.**)

(ii) Note that $\tilde{\Phi}_X$ and its eigenvectors are supported in the subspace of $(\mathbb{C}^{d+1})^{\otimes |X|}$ spanned by $|i_1 \cdots i_{|X|}\rangle$ for which exactly one if the i_p 's is non-zero. Therefore $|0 \cdots 0\rangle$ is orthogonal to all eigenvectors of $\tilde{\Phi}_X$ and thus $|0 \cdots 0\rangle$ is an eigenvector of Γ_X with eigenvalue $1 - \text{Tr}(H_X)$.

(iii) The partial trace of $\Phi_{[k]}$ over $[k] \setminus X$, when performed in the computational basis $|i_1 \cdots i_k\rangle$, can be divided into three portions depending on the number of non-zero entries in $(i_p \mid p \in [k] \setminus X)$; first, is all indices i_p are zero, second, where exactly one i_p is non-zero, and third where more than one index i_p is non-zero. The respective contributions are as follows:

$$\text{Tr}_{[k] \setminus X}(\Phi_{[k]}) = \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + 0 \quad (182)$$

Recalling the definition of $\Gamma_{[k]}$,

$$\Gamma_{[k]} := \Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]} \quad (183)$$

One concludes

$$\begin{aligned} \text{Tr}_{[k] \setminus X}(\Gamma_{[k]}) &= \text{Tr}_{[k] \setminus X}(\Phi_{[k]} + (1 - \text{Tr}(H_{[k]}))Z_{[k]}), \\ &= \tilde{\Phi}_X + \text{Tr}(H_{[k] \setminus X})Z_X + (1 - \text{Tr}(H_{[k]}))Z_X, \\ &= \tilde{\Phi}_X + (1 - \text{Tr}(H_X))Z_X, \\ &= \Gamma_X. \end{aligned} \quad (184)$$

□

Corollary 27. By Lemma 26, one has $\text{Tr}(\Gamma_X) = 1$. If the eigenvalues of H_X satisfy

$$\text{Tr}(H_X) \leq 1, \quad (185)$$

then Γ_X is positive semidefinite, and thus Γ_X is a density matrix. In such a case, assuming $H_X \neq 0$ so that $\text{Tr}(H_X) \in (0, 1]$, the von Neumann entropy of Γ_X is related to the von Neumann entropy of $H_X / \text{Tr}(H_X)$ by the equation⁸

$$S(\Gamma_X) = h(\text{Tr}(H_X)) + \text{Tr}(H_X) S\left(\frac{H_X}{\text{Tr}(H_X)}\right). \quad (186)$$

Moreover, if the eigenvalues of H_X satisfy the stronger condition

$$\lambda_1(H_X) + \text{Tr}(H_X) \leq 1, \quad (187)$$

then the maximum eigenvalue of Γ_X is $1 - \text{Tr}(H_X)$.

Proof. The only non-trivial claim involves the relation between the von Neumann entropies of Γ_X and $H_X / \text{Tr}(H_X)$, respectively given by

$$S(\Gamma_X) = -(1 - \text{Tr}(H_X)) \log(1 - \text{Tr}(H_X)) - \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X), \quad (188)$$

$$S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = -\frac{1}{\text{Tr}(H_X)} \sum_{i=1}^d \lambda_i(H_X) \log \lambda_i(H_X) + \log(\text{Tr}(H_X)). \quad (189)$$

⁸Here $h(p) = -p \log p - (1 - p) \log(1 - p)$ denotes the binary entropy of $p \in [0, 1]$.

Multiplying the von Neumann entropy of $H_X/\text{Tr}(H_X)$ by $\text{Tr}(H_X)$ yields

$$\text{Tr}(H_X)S\left(\frac{H_X}{\text{Tr}(H_X)}\right) = S(\Gamma_X) + \text{Tr}(H_X)\log(\text{Tr}(H_X)) + (1 - \text{Tr}(H_X))\log(1 - \text{Tr}(H_X)), \quad (190)$$

which is the claimed result. $\square$

4.2 The $U(d)$ -recoupling operator as a S_d -recoupling operator

[sec:recoupling-correspondence](#)

The purpose of this section is to describe how to view certain aspects of the recoupling theory for the unitary group $U(d)$ as a special case of the recoupling theory for the symmetric group S_d , at least when d is taken to be sufficiently large. This connection between $U(d)$ and S_d is derived from considering the symmetric group S_d as a *subgroup* of $U(d)$ with respect to the representation of S_d as $d \times d$ permutation matrices. When considering the symmetric group, S_d , as a subgroup of the unitary group, $U(d)$, in this manner, Nishiyama describes the restriction of an irreducible representation of $U(d)$ to the symmetric group in terms of its decomposition into irreducible representations of S_d [Nis00, Theorem 2.3].

Theorem 28. *Let $d \in \mathbb{N}$ and let $\lambda \vdash d$. Let V_λ denote the irrep of $U(d)$ with highest weight λ and W_λ the irrep of S_d . Given a partition μ with length $\ell(\mu) \leq d$, let Υ_μ denote the Young subgroup $\Upsilon_\mu = S_{\mu_1} \times \cdots \times S_{\mu_{\ell(\mu)}} \subseteq S_d$ and let $\tilde{\mu}$ denote the (not necessarily sorted) partition of $\ell(\mu)$ corresponding to the multiplicities of μ . Alternatively, $\tilde{\mu}$ can be characterised in terms of the normaliser of Υ_μ inside S_d via the formula*

$$N_{S_d}(\Upsilon_\mu) = \Upsilon_{\tilde{\mu}} \ltimes \Upsilon_\mu \quad (191)$$

Then the restriction of the irrep V_λ of $U(d)$ to the symmetric group S_d decomposes as follows:

$$\text{Res}_{S_d}^{U(d)} V_\lambda \simeq \bigoplus_{\mu \vdash_d |\lambda|} \text{Ind}_{S_{d-\ell(\mu)} \times \Upsilon_{\tilde{\mu}}}^{S_d} (1 \otimes W_\lambda^{\Upsilon_\mu}) \quad (192)$$

[eq:nishiyama-decomp](#)

where 1 denotes the trivial representation of $S_{d-\ell(\mu)}$ and $W_\lambda^{\Upsilon_\mu}$ is the Υ_μ -invariant subspace of W_λ supporting a representation of $\Upsilon_{\tilde{\mu}}$.

Among the irreducible representations of S_d appearing with non-zero multiplicity in the decomposition of $\text{Res}_{S_d}^{U(d)} V_\lambda$, those labelled by Young diagrams with the largest possible first part are of particular interest [CŞW18].

Definition 7. Let $\lambda = (\lambda_1, \lambda_2, \dots, \lambda_{\ell(\lambda)})$ be a Young diagram of size $|\lambda|$. Let $d \geq \lambda_1 + |\lambda|$ and define the (size d) *augmentation* of λ as

$$\lambda[d] := (d - |\lambda|, \lambda_1, \lambda_2, \dots, \lambda_{\ell(\lambda)}). \quad (193)$$

Note that $\lambda[d]$ also a Young diagram (because $d - |\lambda| \geq \lambda_1$) of size $d = |\lambda[d]|$ and length $\ell(\lambda[d]) = \ell(\lambda) + 1$.

Here we have adopted the notation for $\lambda[d]$ from [Pak22].

The following is [CŞW18, Lemma 13]:

[lem:largest_first_rows](#)

Lemma 29. *Let λ be a positive integer partition and let $d \in \mathbb{N}$ be given. Let V_λ be the irrep of $U(d)$ with highest weight λ and let $\mu \vdash d$ be an arbitrary Young diagram of size d . Define M_μ^λ as the multiplicity space*

$$M_\mu^\lambda := \text{Hom}_{S_d}(W_\mu, \text{Res}_{S_d}^{U(d)} V_\lambda). \quad (194)$$

If M_μ^λ is non-zero, then the first row of μ cannot be smaller than $d - \lambda$:

$$M_\mu^\lambda \neq 0 \implies \mu_1 \geq d - |\lambda|. \quad (195)$$

If $d < \lambda_1 + |\lambda|$, then the above bound is strict

$$M_\mu^\lambda \neq 0 \implies \mu_1 > d - |\lambda|. \quad (196)$$

Otherwise $d \geq \lambda_1 + |\lambda|$ and the above bound is saturated only when $\mu = \lambda[d]$ and moreover $\dim(M_{\lambda[d]}^\lambda) = 1$ in this case.

At this stage, we note that the Littlewood-Richardson coefficients for $U(d)$,

$$c_{\alpha,\beta}^\gamma = \dim \text{Hom}(V_\gamma, V_\alpha \otimes V_\beta), \quad (197)$$

are known to be independent of d (whenever $d \geq \max\{\ell(\alpha), \ell(\beta), \ell(\gamma)\}$), while the Kronecker coefficients

$$g_{\mu,\nu}^\lambda = \dim \text{Hom}(W_\lambda, W_\mu \otimes W_\nu) \quad (198)$$

are implicitly dependent on d because $d = |\mu| = |\nu| = |\lambda|$.

As an immediate corollary of Lemma 29, one obtains a method (originally suggest by Murnaghan [Mur55] and later formalized by Littlewood [Lit58]) for viewing Littlewood-Richardson coefficients as a special types of Kronecker coefficients by appropriately augmenting the Young diagrams involved:

Corollary 30. *Let α, β, γ be Young diagrams and let d be sufficiently large such that $\alpha[d], \beta[d]$ and $\gamma[d]$ are all well-defined. Then the multiplicity space of S_d -intertwiners from $W_{\gamma[d]}$ to $W_{\alpha[d]} \otimes W_{\beta[d]}$ is obtained from the multiplicity space of $U(d)$ -intertwiners from V_γ to $V_\alpha \otimes V_\beta$ by restriction from $U(d)$ to S_d :*

$$C_{\alpha,\beta}^\gamma = \text{Hom}_{U(d)}(V_\gamma, V_\alpha \otimes V_\beta) \simeq \text{Hom}_{S_d}(W_{\gamma[d]}, W_{\alpha[d]} \otimes W_{\beta[d]}) = G_{\alpha[d],\beta[d]}^{\gamma[d]}. \quad (199)$$

In particular, their dimensions are equal: $c_{\alpha,\beta}^\gamma = g_{\alpha[d],\beta[d]}^{\gamma[d]}$.

Proof. The following two representations of S_d are isomorphic (because restriction is a monoidal functor):

$$\text{Res}_{S_d}^{U(d)}(V_\alpha \otimes V_\beta) \simeq \text{Res}_{S_d}^{U(d)}(V_\alpha) \otimes \text{Res}_{S_d}^{U(d)}(V_\beta). \quad \text{\texttt{\{eq:res_monoidal\}}} \quad (200)$$

Applying Lemma 29 (noting d is sufficiently large), yields

$$\text{Res}_{S_d}^{U(d)}(V_\alpha \otimes V_\beta) = \bigoplus_{|\gamma|=|\alpha|+|\beta|} c_{\alpha,\beta}^\gamma \text{Res}_{S_d}^{U(d)}(V_\gamma) = \bigoplus_{\substack{|\gamma|=|\alpha|+|\beta| \\ d-|\gamma| \geq \gamma_1}} c_{\alpha,\beta}^\gamma W_{\gamma[d]} \oplus \cdots, \quad (201)$$

$$\text{Res}_{S_d}^{U(d)}(V_\alpha) \otimes \text{Res}_{S_d}^{U(d)}(V_\beta) = (W_{\alpha[d]} \oplus \cdots) \otimes (W_{\beta[d]} \oplus \cdots) = \bigoplus_{\substack{|\gamma|=|\alpha|+|\beta| \\ d-|\gamma| \geq \gamma_1}} g_{\alpha[d],\beta[d]}^{\gamma[d]} W_{\gamma[d]} \oplus \cdots. \quad (202)$$

In both formulas above, the “...” refers to a direct sum of irreps of S_d labelled by $\lambda \vdash d$ whose first part, λ_1 , is larger than the first part, $d - |\gamma|$, of $\gamma[d]$. By Eq. (200), one concludes there exists an isomorphism between all multiplicity spaces, and in particular, $c_{\alpha,\beta}^\gamma = g_{\alpha[d],\beta[d]}^{\gamma[d]}$. $\square$

Remark 11. Note that this result indirectly implies a stability result for the Kronecker coefficients, namely that the value of $g_{\lambda+(d),\nu+(d)}^{\mu+(d)}$ is eventually independent of d and moreover reaches the value $c_{\lambda_{-1}}^{\mu_{-1},\nu_{-1}}$ where λ_{-1} is obtained from λ by removal of the first row of λ .

cor:ud_sd_recouple_correspondence

Corollary 31. *Let $\alpha, \beta, \gamma, \mu, \nu, \lambda$ be Young diagrams. Furthermore, take d be sufficiently large such that $\alpha[d], \beta[d], \gamma[d], \mu[d], \nu[d], \lambda[d]$ are all well-defined. Then the following recoupling operator for S_d is obtained from the corresponding recoupling operator for $U(d)$ by restriction from $U(d)$ to S_d :*

$$\begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \simeq \begin{bmatrix} \alpha[d] & \beta[d] & \mu[d] \\ \gamma[d] & \lambda[d] & \nu[d] \end{bmatrix}. \quad (203)$$

[Proposed outline as of May 30 2024, for drafting]

Below is the proposed outline of the paper as of May 30, 2024. Our proposed journal is *Communications in Mathematical Physics*. It will be ideal to illustrate with figures.

- introduction (include 2x2 case)
- general definition
- relationship to Horn's problem, (how does it generalize the problem)
- representation theoretic/quantization analogue (Ponzano-Regge and connection to state-sum models? etc.)
- comment about deviation from Thompson's conjecture (correspondence between eigenvalues of sums and logs of singularvalues of products)
- spectral embedding into the quantum marginal problem
- derivation of entropic inequalities (both from rep theory directly, and from embedding into QMP, things like concavity and data processing inequality for Renyi-entropies, etc), then prove they go beyond face constraints

5 [Research notes, for drafting]

- For a comprehensive reference textbook for the $6j$ coefficients, see the textbook by Varshalovich, Moskalev, and Khersonskii [VMK88], specifically Chapter 9.
- A popular textbook on quantum groups with an emphasis on the connection to Poisson manifolds, Lie bialgebras and their quantizations by Chari and Pressley [CP95]
- The general subject of multivariate trace inequalities involving exponentials of sums of Hermitian and products of their exponentials is, of course, intimately related. See for instance the paper by Sutter, Berta, and Tomamichel [SBT17].
- Also the article [FT14] by Forrester and Thompson which describes the connection to hyperbolic geometry quite explicitly.
- Course notes on Poisson geometry with an eye toward quantization by Ciccoli and Witkowski [CW06]
- An excellent expository overview of the big picture by Knutson [Knu00]
- The masters essay/thesis of Backens (TODO: check if the pdf Matthias sent to me is the final version) [Bac10]
- An excellent introduction to these ideas is the recent masters thesis on various aspects of additive and multiplicative Horn problems in connection to Vinnikov curves and the Lax conjecture due to Hess and supervised by S. Kuhlmann and M. Infusino [Hes20]
- The PhD thesis of Esposito (from the maths department here in Copenhagen) serves as a introduction to the world of Poisson momentum maps and the notion of a “quantum” momentum map as a deformation of the ordinary “classical” momentum map (where deformation is in the sense of deformation quantization) [Esp12].
- Alekseev [Ale97]
- Papers contained results pertaining to the generalization of Duistermaat-Heckman measures for Hamiltonian group actions with group-valued moment maps by Alekseev, Meinrenken, and Woodward [AMW99; AMW00]
- The proof of Thompson’s conjecture in the style of Alekseev, Meinrenken, and Woodward is the “linearization paper” due to [AMW01].
- See also the related article due to Lamb which summarizes the ideas of [AMW01] while also providing more technical background [Lam05]
- Lu and Evens extend the proof technique of [AMW01] to a larger class of Lie groups [LE03]
- Alekseev, Podkopaeva, and Szenes [APS17a]
- Alekseev, Podkopaeva, and Szenes [APS17b]
- Alekseev and Meinrenken [AM07]
- Deforming the symplectic volume by the s parameter Alekseev et al. [Ale+18]
- Klyachko [Kly00]
- Christandl [Chr08]
- Christandl, Şahinoğlu, and Walter [CŞW18]
- Kirillov and Reshetikhin [KR90]

- Usage of Wigner’s $6j$ symbol to compute asymptotic errors rates for the promise problem of distinguishing between two possible “change points” for a sequence of unknown pure states to transition from one unknown type to another [AH11]
- A great summary by Roberts [Rob02]; also the original result for the asymptotics of the $6j$ symbols due to Roberts [Rob99]
- Coquereaux and Zuber explicitly outline the link between the orbital measures (i.e. the induced distributions in the sense of Klyachko’s random walk approach) and the magnitude of the stretched Littlewood-Richardson coefficients [CZ18]. See also the papers [CMZ20; CMZ19; Zub18].
- A recent article by McSwiggen is an excellent introduction to the Harish-Chandra integral [McS21]. The proof of the Harish-Chandra formula provide by McSwiggen using the heat equation methods from Fegan’s 1978 paper [Feg78] seems very promising for deriving a connection to the character formula for the norm of the recoupling operator!
- See also Coquereaux’s later paper which promotes the alternative, albeit equivalent, usage of the integer-valued gon and tet functions in place of the rational-valued theta (triangular) and irrational-valued $6j$ symbols [Coq23]. Most relevant for this project is [Coq23, Section IV D] which discusses the apparent lack of substantial generalization from $SU(2)$ to $SU(N)$ (“*apart from very few special cases (low dimension of irreps and no multiplicity) worked out by physicists at the time of the eightfold way, we are not aware of any published general result*”).
- The recent paper by Alekseev, Morozov, and Sleptsov (different Alekseev!) covers the case of some multiplicity free formulas for the quantum $6j$ symbols for $U_q(\mathfrak{sl}_N)$ [AMS20]
- Keppeler, Plätzer, and Sjodahl recently calculated $SU(d)$ $6j$ symbols with adjoint representations [KPS24] and a bit earlier [Alc+23] calculated $SU(d)$ $6j$ symbols with at least two fundamental representations [Alc+23].
- Boalch [Boa07]
- Taylor and Woodward [TW06]
- Henriques and Kamnitzer [HK05]
- Octahedron Recurrence/Rule Robbins and Rumsey Jr [RR86] applied to the “Hive” ring Knutson, Tao, and Woodward [KTW03], and also Henriques and Kamnitzer [HK06]
- The integral formula for the square of the $SU(2)$ $6j$ symbol can be performed by an appropriate change of variables [FL03]. See also other papers by Laurent Freidel and collaborators that are relevant [FK00; FNR07; FKL10].
- Speyer [Spe05]
- Generalizations of Horn’s problem to the product of unitary matrices [AW97; Bel08; BK16]
- Bhatia [Bha01]
- History account offered by Daftuar and Hayden [DH05], and technical review article by Fulton [Ful00].
- Connecting group-valued moment maps to spherical, flat, and hyperbolic tetrahedron via Minkowski’s theorem from the reconstruction of flat tetrahedron Haggard, Han, and Riello. See also the comments in [HHR16, Section 11] on the utility of *quasi-poisson* (controlled failure of the Jacobi identity) manifolds and group-valued moment maps.
- Foundational results concerning the relationships between the edge lengths, face areas, dihedral and solid angles of a tetrahedron by Richardson [Ric02]
- Potential application to inverse kinematics of 3-link arms in three dimensions? (Thanks Harold!)

- **TODO:** track down pdfs for the original triple of papers by Thompson and Therianos titled “On the singular values of matrix products” I, II, and III from 1973
- **TODO:** track down pdfs for the original triple of papers by Thompson and Freede titled “On the eigenvalues of sums of Hermitian matrices” I, II, and III from 1973
- **TODO:** track down a pdf for Thompson’s 1988 “Matrix Spectral Inequalities” referenced by Klyachko
- Thompson and Therianos [TT73] present a number of interesting counter-examples to the question of whether the inequalities pertaining to the spectra of general matrices can be deduced from looking at the spectra of diagonal matrices (both in the additive and multiplicative cases).
- Elementary introduction to toy models of three-dimensional quantum space [Bar03]
- Horn’s original paper where they conjectured the form of a necessary and sufficient set of inequalities via an inductive scheme for increasing dimension [Hor62]
- Berline, Vergne, and Walter a pedagogical account of the triple role played by Horn’s inequalities in characterizing the existence of invariants vectors in tensor products, the eigenvalues of sums of Hermitian matrices, and intersections of Schubert varieties [BVW18]
- Regge and Williams [RW00] a relatively recent review of discrete approaches to gravity. The essential idea, in contrast to the differential geometry of general relativity, is to model curvature discretely instead of continuously. In particular, one can think of a triangulation of the surface of a sphere; the curvature of the sphere is encoded in the triangulation when a configuration of three (or more) triangles meet at a common vertex with sum of interior angles deviating from π . It is in this sense that Regge and Williams refers to these types of models as “piecewise-flat”. Contained is an overview of various attempts to generalize discrete theories of three-dimensional gravity to four dimensions [RW00, Section III].
- Fulton provides a comprehensive treatment of the connection between representation theory of $GL(d)$ and the geometry and (co)homology of the complex Grassmannians [Ful97]
- Alcock-Zelinger consider the case of $6j$ symbols for $SU(N)$ for wherein two opposing edges are labelled by fundamental irreps of $SU(N)$ so that all vertices are multiplicity one (or zero) [Alc18] **TC:** They also freely use the birdtrack notation which I enjoy.
- Only the tetrahedral symmetry of the $U_q(SL(N))$ $6j$ -symbol is known, the counterpart of the Regge symmetry is still unknown, although Morozov and Sleptsov conjectures the generalized Regge symmetry may be related to the eigenvalue conjecture, just as it is in the case when $N = 2$ [MS18].
- The graphical notion, specifically the use of an additional edge for multiplicity spaces can has been used for a long time, as early as [AB68; Mou84].
- The original Ponzano-Regge paper [PR68]
- Gurau derives the Ponzano-Regge asymptotic formula directly using Stirling’s approximation applied to Racah’s single-sum formula [Gur08]
- Bartlett and Nzaganya work with so-called *coherent loop states* or *Bohr-Sommerfeld states* to develop some semiclassical results concerning the inner products between such states (see also the talk on Youtube) [BN23]
- Bianchi, Dona, and Speziale uses Minkowski’s theorem to parametrize convex polyhedra in $\mathbb{R}^3$ with specified face areas and the associated Kapovich-Millson phase space (i.e., the *space of shapes of polyhedra with fixed areas* and then considers its connection to loop quantum gravity via the usual Guillemin-Sternberg quantization commutes with reduction philosophy which associates invariants of the quantization of the unreduced phase space to the reduced phase space [BDS11] (notice the connection with [HHR16]).

- Aquilanti et al. provides an excellent introduction to subject of classical $6j$ symbols, their history, diagrammatics, presentations and asymptotics [Aqu+12]
- Marzuoli and Rasetti connects quantum computing models to the recoupling theory of $SU(2)$ angular momentum and also considers the semiclassical limit [MR05]
- Charles quantization of polygon spaces, tetrahedron and semiclassical limits [Cha10]
- Regarding an algorithmic construction for the polyhedra appearing in Minkowski's theorem and their complexity, see this post: <https://math.stackexchange.com/questions/105033/constructive-algorithm-for-mi>
- Artamonov derives an explicit formula for the $d = 3$ case [Art23], and later to arbitrary d [Art24].
- There is a series of papers involving Alexey Sleptsov, starting with a computation of quantum $6j$ symbols for generic dimension d but with symmetric representations in terms of certain hypergeometric polynomials [MMS17], then with a discussion of the eigenvalue hypothesis and its potential connection to generalized Regge symmetries [MS18], then with a calculation of all multiplicity free quantum $6j$ symbols for arbitrary dimension [AMS20], and then finally with the discovery of novel “tug-the-hook” symmetries for quantum $6j$ symbols for all d [LS23].
- Prakash Belkale has contributed greatly to the theory surrounding Horn-type problems: for example, in [Bel04], Belkale provides a *geometric* proof of the equivalence between (i) the structure coefficients of the representation theory of $GL(d)$ (i.e. the LR-coefficients $c_{\mu\nu}^\lambda$), and (ii) the structure coefficients of the cohomology ring of the Grassmannian $Gr(d, n)$ of d -dimensional subspaces in $\mathbb{C}^d$. This proof is now complementary to the previously known algebraic proofs which rely on knowledge of the algebra of symmetric functions. This geometric approach taken by Belkale, which emphasizes the tangent spaces of the involved Schubert varieties, is later used by Belkale to provide a geometric proof of the saturation conjecture [Bel02].

A Proof of symmetric group recoupling operator upper bound

sec:sk_recouple_bound_proof

The purpose of this section is to prove ??.

TODO: Make the notation and diagrammatic conventions used here match the rest of the document, also resolve ambiguities introduced by the the “dot” notation.

Definition 8 (S_k recoupling operators). Given six Young diagrams $\alpha, \beta, \gamma, \mu, \nu$ and λ (each with fixed size $k \in \mathbb{N}$), the *recoupling operator* is a linear map

$$\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix} : G_{\alpha,\beta}^{\mu} \otimes G_{\mu,\gamma}^{\lambda} \rightarrow G_{\alpha,\nu}^{\lambda} \otimes G_{\beta,\gamma}^{\nu}, \quad (204)$$

between the multiplicity spaces of irreps of the symmetric group defined as

$$\begin{array}{c} G_{\alpha,\nu}^{\lambda} \uparrow \quad G_{\beta,\gamma}^{\nu} \uparrow \\ \boxed{\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix}} \\ G_{\alpha,\beta}^{\mu} \uparrow \quad G_{\mu,\gamma}^{\lambda} \uparrow \end{array} := \frac{1}{\textcircled{V_{\lambda}}} \begin{array}{c} G_{\alpha,\nu}^{\lambda} \swarrow \quad G_{\beta,\gamma}^{\nu} \searrow \\ \begin{array}{ccc} & V_{\nu} & \\ V_{\alpha} \nearrow & V_{\beta} & \nwarrow \\ & V_{\gamma} & \\ G_{\alpha,\beta}^{\mu} \swarrow & G_{\mu,\gamma}^{\lambda} \searrow & \end{array} \end{array} \quad (205)$$

The recoupling operator arises when applying Schur’s lemma (with multiplicities) to diagram on the left-hand side of the subsequent equation:

$$\begin{array}{c} \lambda \quad M_{\lambda}^{\alpha,\nu} \quad \nu \quad M_{\nu}^{\beta,\gamma} \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \alpha \quad \beta \quad \gamma \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \lambda \quad M_{\mu}^{\alpha,\beta} \quad \mu \quad M_{\lambda}^{\mu,\gamma} \end{array} = \begin{array}{c} M_{\lambda}^{\alpha,\nu} \quad M_{\nu}^{\beta,\gamma} \\ \uparrow \quad \uparrow \\ \boxed{\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix}} \\ \downarrow \quad \downarrow \\ \lambda \quad M_{\mu}^{\alpha,\beta} \quad M_{\lambda}^{\mu,\gamma} \end{array} \quad (206)$$

The recoupling operator derives its name from the recoupling theory of angular momenta and serves the purpose of a change of basis between two distinct (de-)coupling schemes. Specifically, the recoupling operator satisfies the following equation:

$$\begin{array}{c} \alpha \quad \beta \quad \gamma \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \mu \quad \lambda \end{array} = \sum_{\nu} M_{\lambda}^{\alpha,\nu} M_{\nu}^{\beta,\gamma} \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \mu \quad \lambda \quad \nu \end{array} \quad (207)$$

{eq:recoupling-trees}

As a shorthand for this equation, we simply write:

$$\begin{array}{c} \alpha \quad \beta \quad \gamma \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \mu \quad \lambda \end{array} \quad \begin{array}{c} \boxed{\begin{bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{bmatrix}} \\ \text{-----} \end{array} \quad \begin{array}{c} \alpha \quad \beta \quad \gamma \\ \swarrow \quad \nearrow \quad \searrow \quad \nearrow \\ \lambda \quad \nu \end{array} \quad (208)$$

The next result is a consequence of an application of Schur's lemma to the identity on W_α :

$$\begin{array}{c} \alpha \\ \downarrow \\ \text{Diagram 1} \\ \uparrow \\ \alpha \end{array} M_\mu^{\alpha,\beta} = \frac{1}{\text{Diagram 2}} = \frac{\text{Diagram 3}}{\text{Diagram 4}} = \begin{array}{c} \alpha \\ \downarrow \\ \text{Diagram 5} \\ \uparrow \\ \alpha \end{array} M_\mu^{\alpha,\beta}. \quad (209)$$

Combining two instances of the above result yields:

$$\begin{array}{c} M_\mu^{\alpha,\beta} \quad M_\nu^{\beta,\gamma} \\ \downarrow \quad \downarrow \\ \text{Diagram 1} \\ \uparrow \quad \uparrow \\ M_\mu^{\alpha,\beta} \quad M_\nu^{\beta,\gamma} \end{array} = \frac{\text{Diagram 2}}{\text{Diagram 3}} = \begin{array}{c} M_\mu^{\alpha,\beta} \quad M_\nu^{\beta,\gamma} \\ \downarrow \quad \downarrow \\ \text{Diagram 4} \\ \uparrow \quad \uparrow \\ M_\mu^{\alpha,\beta} \quad M_\nu^{\beta,\gamma} \end{array}. \quad (210)$$

Proof of ??. Applying the Cauchy-Schwarz inequality along a particular edge-cut indicated below, and applying Schur's lemma multiple times yields

$$\begin{aligned} & \left(\text{Diagram 1} \right)^2 = \left(\text{Diagram 2} \right) = \left(\text{Diagram 3} \right) \\ & \leq \left[\text{Diagram 4} \right]^{\frac{1}{2}} = \frac{1}{\left(\text{Diagram 5} \right)} \left[\text{Diagram 6} \right]^{\frac{1}{2}} \\ & = \frac{\left(\text{Diagram 7} \right)^2}{\left(\text{Diagram 8} \right)} \left[\text{Diagram 9} \right]^{\frac{1}{2}} \end{aligned} \quad (211)$$



B Proof of recoupling operator norm equivalences

[sec:recouple_norms](#)

TODO: Figure out if a “Young recoupling operator” is even a useful object to consider now that my entropy inequality proof failed.

[defn:young_recouple](#)

Definition 9. Let $a, b, c \in \mathbb{N}_{>0}$ be given and let

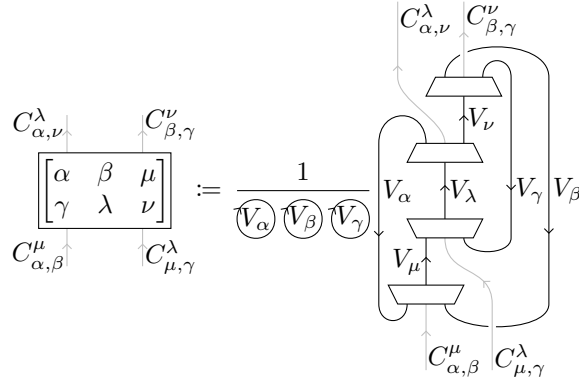
$$\alpha \in \Upsilon_{a,\infty}, \quad \beta \in \Upsilon_{b,\infty}, \quad \gamma \in \Upsilon_{c,\infty}, \quad \mu \in \Upsilon_{a+b,\infty}, \quad \nu \in \Upsilon_{b+c,\infty}, \quad \lambda \in \Upsilon_{a+b+c,\infty}. \quad (212)$$

The *Young recoupling operator* is the operator

$$\left\{ \begin{array}{ccc} \alpha & \beta & \gamma \\ \delta & \epsilon & \phi \end{array} \right\} : C_{\alpha,\beta}^\mu \otimes C_{\mu,\gamma}^\lambda \rightarrow C_{\alpha,\nu}^\lambda \otimes C_{\beta,\gamma}^\nu, \quad (213)$$

between the multiplicity spaces $C_{\alpha,\beta}^\mu := \text{Hom}_{S_a \times S_b}(W_\alpha \otimes W_\beta, W_\mu)$ defined as

$$\left[\begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right] := \frac{\text{Tr}_{W_\alpha \otimes W_\beta \otimes W_\gamma}[(\text{id}_{W_\alpha} \otimes Y_{\beta\gamma}^{\nu;\dagger}) Y_{\alpha\nu}^{\lambda;\dagger} Y_{\mu\gamma}^\lambda (Y_{\alpha\beta}^\mu \otimes \text{id}_{W_\gamma})]}{\dim(W_\alpha \otimes W_\beta \otimes W_\gamma)} \quad (214)$$



$$\left[\begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right] := \frac{1}{V_\alpha V_\beta V_\gamma} \quad (215)$$

Lemma 32. Let $\left\{ \begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right\}$ be the recoupling operator for $U(d)$ and let $\left[\begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right]$ be the Young recoupling operator. Then

$$\|\Pi_\nu \Pi_\mu \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\lambda\|_\infty = \left\| \left\{ \begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right\} \right\|_\infty = \left\| \left[\begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right] \right\|_\infty \quad (216)$$

[eq:recoupling_prod_proj_iso_norm](#)

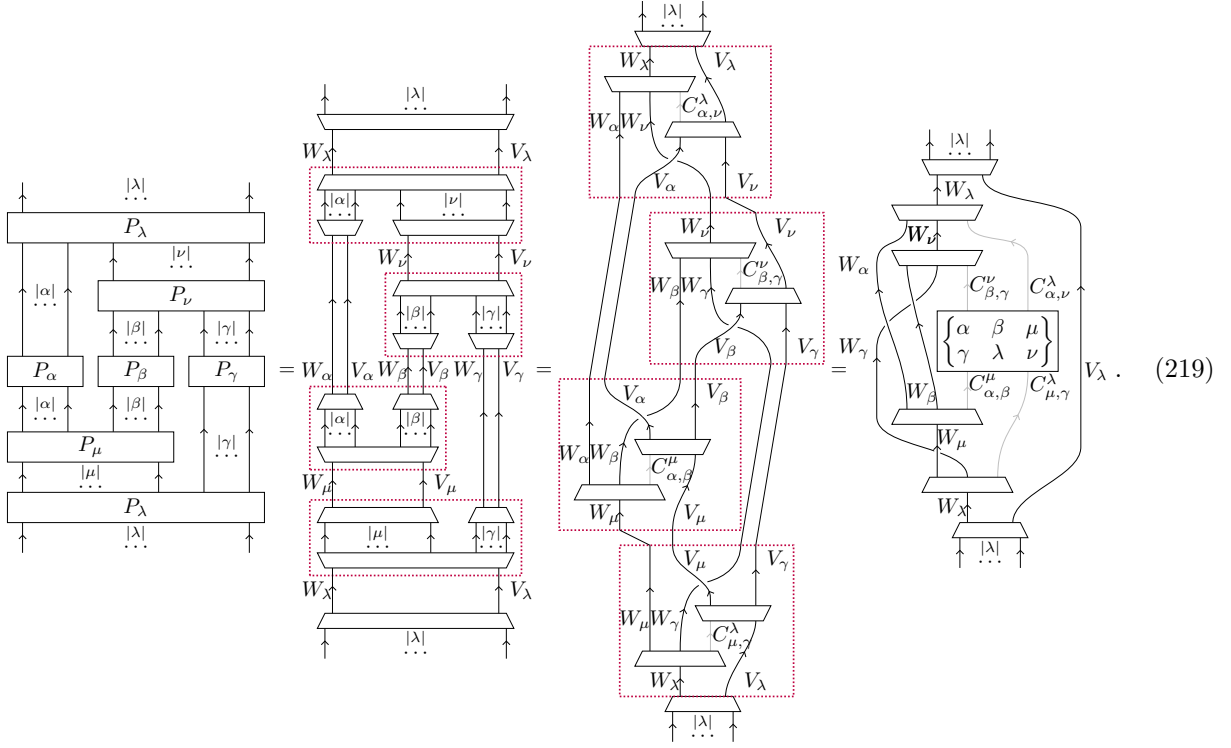
Proof. The claim follows by demonstrating the existence of isometries $\iota_I, \iota_{II}, \iota_{III}$, and ι_{IV} such that

$$\Pi_\nu \Pi_\mu \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\lambda = \iota_{II} \left(\text{id}_{V_\lambda} \otimes \text{id}_{W_\alpha} \otimes \text{id}_{W_\beta} \otimes \text{id}_{W_\gamma} \otimes \left\{ \begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right\} \right) \iota_I^\dagger, \quad (217)$$

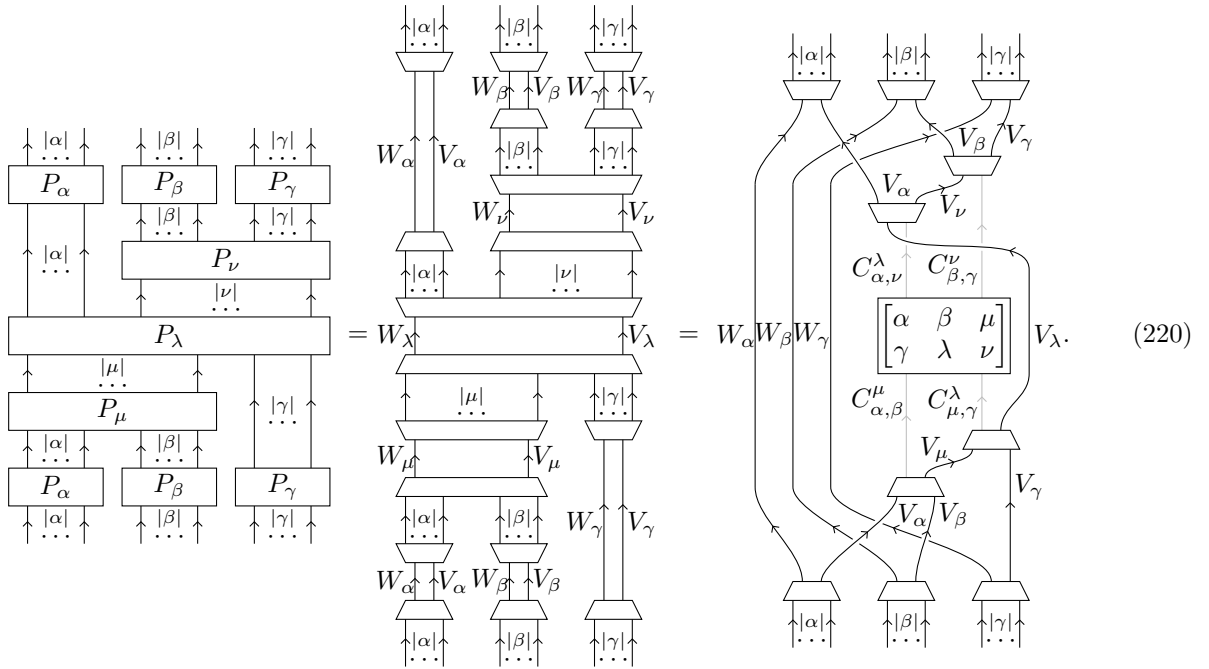
$$= \iota_{III} \left(\text{id}_{V_\lambda} \otimes \text{id}_{W_\alpha} \otimes \text{id}_{W_\beta} \otimes \text{id}_{W_\gamma} \otimes \left[\begin{array}{ccc} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{array} \right] \right) \iota_{IV}^\dagger. \quad (218)$$

The first two isometries, ι_I and ι_{II} , involve decompositions of tensor products of irreps of the symmetric

group:



The remaining two isometries, ι_{III} and ι_{IV} , involve decompositions of tensor products of irreps of the unitary group:



□

C [Calculations from previous draft]

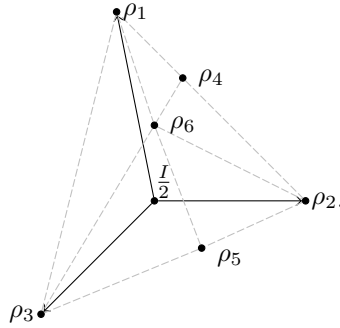
C.1 From states to recoupling

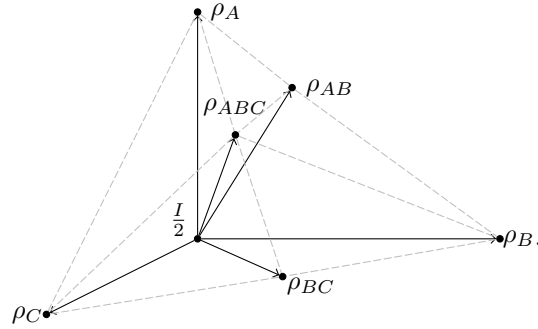
In this section we repeat the proof technique used in [Chr08] to prove the existence of a sextuple of convergent sequences of Young diagrams satisfying the recoupling operator lower bound.

Let $(p_1, p_2, p_3) \in \Delta_3$, let $\rho_1, \rho_2, \rho_3, \rho_4, \rho_5, \rho_6$ be a sextuple of $d \times d$ density operators (with spectra $r_i = \text{spec}(\rho_i)$), and suppose that the density matrices $\{\rho_1, \dots, \rho_6\}$ are subject to the constraints

$$\begin{aligned} (p_1 + p_2)\rho_4 &= p_1\rho_1 + p_2\rho_2, \\ (p_2 + p_3)\rho_5 &= p_2\rho_2 + p_3\rho_3, \\ (p_1 + p_2 + p_3)\rho_6 &= p_1\rho_1 + p_2\rho_2 + p_3\rho_3. \end{aligned} \tag{eq:sextuple_config_constraints} \tag{221}$$

In the case of $d = 2$, this configuration can be depicted faithfully as a configuration of points in $\mathbb{R}^3$:


(222)


(223)

Using spectra estimation theorem, we have for each $i \in \{1, \dots, 6\}$, $k_i \in \mathbb{N}$, and $\lambda \vdash k_i$,

$$\text{Tr}(P_\lambda \rho_i^{\otimes k_i}) \leq (k_i + 1)^{d(d-1)/2} \exp(-k_i D(\bar{\lambda} \| r_i)). \tag{224}$$

In particular, if $D(\bar{\lambda} \| r_i) < \epsilon$, then

$$\text{Tr}(P_\lambda \rho_i^{\otimes k_i}) \leq (k_i + 1)^{d(d-1)/2} \exp(-k_i \epsilon). \tag{225}$$

Furthermore, the ensemble approximation lemma applied to $\rho_4 = x\rho_1 + (1-x)\rho_2$ with $x = \frac{p_1}{p_1+p_2}$ and $\mu \vdash k_1 + k_2 = k_4$ (satisfying $k_4 x \approx k_1$ and $k_4(1-x) \approx k_2$)

$$\text{Tr}(P_\mu \rho_4^{\otimes k_4}) = \text{Tr}(P_\mu (x\rho_1 + (1-x)\rho_2)^{\otimes k_4}) \geq \frac{1}{k_4 + 1} \text{Tr}(P_\mu \rho_1^{\otimes k_1} \otimes \rho_2^{\otimes k_2}) \tag{226}$$

Combining this with spectral estimation yields the following bound

$$\text{Tr}(P_\mu \rho_1^{\otimes k_1} \otimes \rho_2^{\otimes k_2}) \leq (k_4 + 1) \text{Tr}(P_\mu \rho_4^{\otimes k_4}) \leq (k_4 + 1)^{d(d-1)/2+1} \exp(-k_4 D(\bar{\mu} \| r_4)). \tag{227}$$

Let $k_6 = k_1 + k_2 + k_4$ and note that $k_i \leq k_6$ for all i . Then one obtains the following collection of upper bounds

$$\mathrm{Tr}(P_\alpha \rho_1^{\otimes k_1}) \leq (k_1 + 1)^{d(d-1)/2} \exp(-k_1 D(\bar{\alpha} \| r_1)), \quad (228)$$

$$\mathrm{Tr}(P_\beta \rho_2^{\otimes k_2}) \leq (k_2 + 1)^{d(d-1)/2} \exp(-k_2 D(\bar{\beta} \| r_2)), \quad (229)$$

$$\mathrm{Tr}(P_\gamma \rho_3^{\otimes k_3}) \leq (k_3 + 1)^{d(d-1)/2} \exp(-k_3 D(\bar{\gamma} \| r_3)), \quad (230)$$

$$\mathrm{Tr}(P_\mu \rho_1^{\otimes k_1} \otimes \rho_2^{\otimes k_2}) \leq (k_4 + 1)^{d(d-1)/2+1} \exp(-k_4 D(\bar{\mu} \| r_4)), \quad (231)$$

$$\mathrm{Tr}(P_\nu \rho_2^{\otimes k_2} \otimes \rho_3^{\otimes k_3}) \leq (k_5 + 1)^{d(d-1)/2+1} \exp(-k_5 D(\bar{\nu} \| r_5)), \quad (232)$$

$$\mathrm{Tr}(P_\lambda \rho_1^{\otimes k_1} \otimes \rho_2^{\otimes k_2} \otimes \rho_3^{\otimes k_3}) \leq (k_6 + 1)^{d(d-1)/2+2} \exp(-k_6 D(\bar{\lambda} \| r_6)). \quad (233)$$

At the moment, there is an important subtlety which has NOT been addressed. In order to apply the ensemble approximation lemma it must be the case that the vales of k_1, k_2 , and k_3 are reasonably good approximations of p_1, p_2 , and p_3 in the sense that the multinomial probability $m_p(k_1, k_2, k_3)$ is taken larger than the crude, albeit polynomial, lower bound of $(k_1 + k_2 + k_3 + 1)^{-2}$. Additionally, in order to apply the ensemble approximation lemma in parallel three times as we have done here, one must show that these reasonably good approximations can be made for (p_1, p_2, p_3) , $(p_1, p_2)/(p_1 + p_2)$ and $(p_2, p_3)/(p_2 + p_3)$ for the *same* tuple of integers (k_1, k_2, k_3) . While this is not too difficult to demonstrate, we omit a proof of this claim here.

At this stage, consider the following sextuple of projection operators actings on $\mathcal{H}^{\otimes k_6}$,

$$\Pi_\alpha := P_\alpha \otimes I^{\otimes k_2 + k_3}, \quad (234)$$

$$\Pi_\beta := I^{\otimes k_1} \otimes P_\beta \otimes I^{\otimes k_3}, \quad (235)$$

$$\Pi_\gamma := I^{\otimes k_1 + k_2} \otimes P_\gamma, \quad (236)$$

$$\Pi_\mu := P_\mu \otimes I^{\otimes k_3}, \quad (237)$$

$$\Pi_\nu := I^{\otimes k_1} \otimes P_\nu, \quad (238)$$

$$\Pi_\lambda := P_\lambda. \quad (239)$$

It can be shown (using applications of Schur's lemma for $U(d)$ for example) that almost all pairs of projection operators in $\{\Pi_\alpha, \Pi_\beta, \dots, \Pi_\lambda\}$ are commuting; the only exception is the pair $\{\Pi_\mu, \Pi_\nu\}$. Define the following projection operators,

$$\Pi_{\alpha\beta\gamma} = \Pi_\alpha \Pi_\beta \Pi_\gamma, \quad (240)$$

$$\Pi_{\alpha\beta\gamma\lambda} = \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\lambda, \quad (241)$$

$$\Pi_{\alpha\beta\gamma\lambda}^\mu = \Pi_\mu \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\lambda, \quad (242)$$

$$\Pi_{\alpha\beta\gamma\lambda}^\nu = \Pi_\nu \Pi_\alpha \Pi_\beta \Pi_\gamma \Pi_\lambda. \quad (243)$$

Additionally, define

$$\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} := \rho_1^{\otimes k_1} \otimes \rho_2^{\otimes k_2} \otimes \rho_3^{\otimes k_3}. \quad (244)$$

In any case, the upper bounds above are equivalent to the following upper bounds:

$$\mathrm{Tr}(\Pi_\alpha \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_1 + 1)^{d(d-1)/2} \exp(-k_1 D(\bar{\alpha} \| r_1)), \quad (245)$$

$$\mathrm{Tr}(\Pi_\beta \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_2 + 1)^{d(d-1)/2} \exp(-k_2 D(\bar{\beta} \| r_2)), \quad (246)$$

$$\mathrm{Tr}(\Pi_\gamma \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_3 + 1)^{d(d-1)/2} \exp(-k_3 D(\bar{\gamma} \| r_3)), \quad (247)$$

$$\mathrm{Tr}(\Pi_\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_4 + 1)^{d(d-1)/2+1} \exp(-k_4 D(\bar{\mu} \| r_4)), \quad (248)$$

$$\mathrm{Tr}(\Pi_\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_5 + 1)^{d(d-1)/2+1} \exp(-k_5 D(\bar{\nu} \| r_5)), \quad (249)$$

$$\mathrm{Tr}(\Pi_\lambda \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq (k_6 + 1)^{d(d-1)/2+2} \exp(-k_6 D(\bar{\lambda} \| r_6)). \quad (250)$$

Henceforth let the polynomial prefactors be accounted for using the expression

$$L_c^d(k) := \left(\frac{d(d-1)}{2} + c \right) \log(k+1). \quad (251)$$

Using $L_c^d(k)$, the above bounds can be re-expressed as

$$\text{Tr}(\Pi_\alpha \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_1 D(\bar{\alpha} \| r_1) + L_0^d(k_1)), \quad (252)$$

$$\text{Tr}(\Pi_\beta \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_2 D(\bar{\beta} \| r_2) + L_0^d(k_2)), \quad (253)$$

$$\text{Tr}(\Pi_\gamma \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_3 D(\bar{\gamma} \| r_3) + L_0^d(k_3)), \quad (254)$$

$$\text{Tr}(\Pi_\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_4 D(\bar{\mu} \| r_4) + L_1^d(k_4)), \quad (255)$$

$$\text{Tr}(\Pi_\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_5 D(\bar{\nu} \| r_5) + L_1^d(k_5)), \quad (256)$$

$$\text{Tr}(\Pi_\lambda \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \leq \exp(-k_6 D(\bar{\lambda} \| r_6) + L_2^d(k_6)). \quad (257)$$

An application of the Cauchy-Schwarz inequality to the Hilbert-Schmidt inner product $\langle X, Y \rangle = \text{Tr}(X^\dagger Y)$ with $X = \Omega^{\frac{1}{2}} \Pi_1$ and $Y = \Omega^{\frac{1}{2}} \Pi_2$ (where Π_i Hermitian projector and Ω positive-semi definite Hermitian) yields

$$|\text{Tr}(\Pi_1 \Omega \Pi_2)|^2 = \left| \text{Tr}((\Omega^{\frac{1}{2}} \Pi_1)^\dagger (\Omega^{\frac{1}{2}} \Pi_2)) \right|^2 \quad (258)$$

$$= \left| \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_2 \rangle \right|^2 \quad (259)$$

$$\leq \langle \Omega^{\frac{1}{2}} \Pi_1, \Omega^{\frac{1}{2}} \Pi_1 \rangle \langle \Omega^{\frac{1}{2}} \Pi_2, \Omega^{\frac{1}{2}} \Pi_2 \rangle \quad (260)$$

$$= \text{Tr}(\Pi_1 \Omega) \text{Tr}(\Pi_2 \Omega) \quad (261)$$

Additionally, if $[\Pi_1, \Pi_2] = 0$ then $|\text{Tr}(\Pi_1 \Omega \Pi_2)| = |\text{Tr}(\Pi_2 \Pi_1 \Pi_2 \Omega)| = \text{Tr}(\Pi_2 \Pi_1 \Pi_2 \Omega) \geq 0$. Similarly, taking $X = (\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \Pi_{\alpha\beta\gamma\lambda}^\mu$ and $Y = (\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \Pi_{\alpha\beta\gamma\lambda}^\nu$ yields

$$\left| \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^\nu) \right|^2 \leq \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}). \quad (262)$$

Chaining these bounds together produces a (perhaps sub-optimal but nevertheless sufficient) upper bound

$$\left| \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^\nu) \right|^4 \quad (263)$$

$$\leq \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^2 \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^2, \quad (264)$$

$$\leq \text{Tr}(\Pi_\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \text{Tr}(\Pi_\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \text{Tr}(\Pi_{\alpha\beta\gamma\lambda} \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^2, \quad (265)$$

$$\leq \text{Tr}(\Pi_\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \text{Tr}(\Pi_\nu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) \text{Tr}(\Pi_\alpha \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \times \dots \quad (266)$$

$$\times \text{Tr}(\Pi_\beta \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \text{Tr}(\Pi_\gamma \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \text{Tr}(\Pi_\lambda \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3})^{\frac{1}{2}} \quad (267)$$

Therefore,

$$\left| \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^\nu) \right| \leq \exp(-k_6 I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} + O(\log(k_6))) \quad (268)$$

where (under the hand-wavy approximations $p_i \approx k_i/k_6$) we define

$$\begin{aligned} I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} &= \frac{p_1}{2} D(\bar{\alpha} \| r_1) + \frac{p_2}{2} D(\bar{\beta} \| r_2) + \frac{p_3}{2} D(\bar{\gamma} \| r_3) + \dots \\ &\dots + \frac{p_1 + p_2}{4} D(\bar{\mu} \| r_4) + \frac{p_2 + p_3}{4} D(\bar{\nu} \| r_5) + \frac{1}{2} D(\bar{\lambda} \| r_6) \end{aligned} \quad (269)$$

The crucial property of the rate function here is that $I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} \geq 0$ always while $I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} = 0$ only if $\bar{\alpha} = r_1, \bar{\beta} = r_2, \dots, \bar{\lambda} = r_6$.

At this stage, one can use the completeness property $\sum_\lambda \Pi_\lambda = I$ to prove

$$1 = \text{Tr}(\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}) = \sum_{\alpha, \beta, \gamma, \mu, \nu, \lambda} \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^\nu). \quad (270)$$

Taking absolute values (and using subadditivity) and applying Eq. (268) produces

$$1 \leq \sum_{\alpha, \beta, \gamma, \mu, \nu, \lambda} \left| \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^{\mu} \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^{\nu}) \right|, \quad (271)$$

$$\leq \sum_{\alpha, \beta, \gamma, \mu, \nu, \lambda} \exp(-k_6 I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} + O(\log(k_6))). \quad (272)$$

As there are at most

$$U(k_1, k_2, k_3) := (k_1 + 1)^{d-1} (k_2 + 1)^{d-1} (k_3 + 1)^{d-1} (k_1 + k_2 + 1)^{d-1} (k_2 + k_3 + 1)^{d-1} (k_1 + k_2 + k_3 + 1)^{d-1} \quad (273)$$

possible summands, the above equation implies there exists a (possibly non-unique) sextuple $\alpha', \beta', \gamma', \mu', \nu', \lambda'$ (implicitly dependent upon the k_i 's) such that

$$\exp(-k_6 I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha', \beta', \gamma', \mu', \nu', \lambda'} + O(\log(k_6))) \geq \left| \text{Tr}(\Pi_{\alpha'\beta'\gamma'\lambda'}^{\mu'} \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha'\beta'\gamma'\lambda'}^{\nu'}) \right| \geq U(k_1, k_2, k_3)^{-1} \quad (274)$$

which implies

$$I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha', \beta', \gamma', \mu', \nu', \lambda'} \leq \frac{1}{k_6} \log U(k_1, k_2, k_3) + \frac{1}{k_6} O(\log(k_6)) = O(\log(k_6)/k_6) \quad (275)$$

which tends to zero as k_i 's tend to infinity. Therefore, this sequence of sextuples of integer partitions converge to the given spectra in the sense that

$$\bar{\alpha}' \rightarrow r_1, \quad \bar{\beta}' \rightarrow r_2, \quad \bar{\gamma}' \rightarrow r_3, \quad \bar{\mu}' \rightarrow r_4, \quad \bar{\nu}' \rightarrow r_5, \quad \bar{\lambda}' \rightarrow r_6. \quad (276)$$

In addition, by considering any purification of $\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}$ one additionally obtains a bound on the *operator norm*, $\|\cdot\|_{\infty}$, of the product of all Π 's (noting that the operator norm also satisfies $\|X \otimes I\|_{\infty} = \|X\|_{\infty}$). This bound takes the form

$$\|\Pi_{\mu}\Pi_{\nu}\Pi_{\alpha}\Pi_{\beta}\Pi_{\gamma}\Pi_{\lambda}\|_{\infty} = \left\| \Pi_{\alpha\beta\gamma\lambda}^{\mu} \Pi_{\alpha\beta\gamma\lambda}^{\nu} \right\|_{\infty} \geq \left| \text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^{\mu} \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^{\nu}) \right|. \quad (277)$$

(The inequality comes from $\|XY\|_{\infty}^2 \geq \langle XYv, XYv \rangle \geq |\langle v, XYv \rangle|^2$ for all v with unit norm.) Furthermore, one can show by a detailed diagrammatic calculation that there exists appropriate isometries ι_I and ι_{II} such that

$$\iota_{II} \left(I_{U_{\lambda}^d} \otimes I_{W_{\alpha}} \otimes I_{W_{\beta}} \otimes I_{W_{\gamma}} \otimes \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right) \iota_I^{\dagger} = \Pi_{\mu}\Pi_{\nu}\Pi_{\alpha}\Pi_{\beta}\Pi_{\gamma}\Pi_{\lambda} \quad (278)$$

which in turn implies an equality of operator norms

$$\left\| \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right\|_{\infty} = \|\Pi_{\mu}\Pi_{\nu}\Pi_{\alpha}\Pi_{\beta}\Pi_{\gamma}\Pi_{\lambda}\|_{\infty} \quad (279)$$

By combining Eqs. (274), (277) and (279) we obtain a inverse polynomial lower bound for the operator norm of the recoupling operator of the form

$$\left\| \begin{Bmatrix} \alpha' & \beta' & \mu' \\ \gamma' & \lambda' & \nu' \end{Bmatrix} \right\|_{\infty} \geq U(k_1, k_2, k_3)^{-1} \geq (k_1 + k_2 + k_3 + 1)^{-6(d-1)} \quad (280)$$

In summary, we have shown that the existence of a certain configuration of density matrices implies a polynomial lower bound on the operator norm of the recoupling operator for $U(d)$ representations (modulo the $p_i \approx k_i/k_6$ approximation which needs to be justified),

Theorem 33. *Let $0 < p_1, p_2, p_3 < 1$ satisfying $p_1 + p_2 + p_3 = 1$ be given and let $(r_1, r_2, \dots, r_6) \in \Delta_d^{1 \times 6}$ be a sextuple of candidate spectra. If there exists a sextuple of density operators $\rho_1, \rho_2, \rho_3, \rho_4, \rho_5$ and ρ_6 with spectra $r_i = \text{spec}(\rho_i)$ satisfying Eq. (221), then there exists a sequence of sextuples $(\alpha^{(n)}, \beta^{(n)}, \gamma^{(n)}, \mu^{(n)}, \nu^{(n)}, \lambda^{(n)})$ of integer partitions satisfying both*

$$\left\| \begin{Bmatrix} \alpha^{(n)} & \beta^{(n)} & \mu^{(n)} \\ \gamma^{(n)} & \lambda^{(n)} & \nu^{(n)} \end{Bmatrix} \right\|_{\infty} \geq (n+1)^{-6(d-1)} \quad (281)$$

(for sufficiently large n) and the following convergences

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{\alpha^{(n)}}{|\alpha^{(n)}|} &= r_1, & \lim_{n \rightarrow \infty} \frac{\beta^{(n)}}{|\beta^{(n)}|} &= r_2, & \lim_{n \rightarrow \infty} \frac{\gamma^{(n)}}{|\gamma^{(n)}|} &= r_3, \\ \lim_{n \rightarrow \infty} \frac{\mu^{(n)}}{|\mu^{(n)}|} &= r_4, & \lim_{n \rightarrow \infty} \frac{\nu^{(n)}}{|\nu^{(n)}|} &= r_5, & \lim_{n \rightarrow \infty} \frac{\lambda^{(n)}}{|\lambda^{(n)}|} &= r_6, \\ \lim_{n \rightarrow \infty} \frac{|\alpha^{(n)}|}{|\lambda^{(n)}|} &= p_1, & \lim_{n \rightarrow \infty} \frac{|\beta^{(n)}|}{|\lambda^{(n)}|} &= p_2, & \lim_{n \rightarrow \infty} \frac{|\gamma^{(n)}|}{|\lambda^{(n)}|} &= p_3. \end{aligned} \tag{282}$$

Remark 12. Note that the satisfaction of Eq. (281) for some $n \in \mathbb{N}$ automatically entails that $\left\{ \frac{\alpha^{(n)}}{\gamma^{(n)}} \frac{\beta^{(n)}}{\lambda^{(n)}} \frac{\mu^{(n)}}{\nu^{(n)}} \right\} \neq 0$ and thus $c_{\alpha,\beta}^\mu, c_{\beta,\gamma}^\nu, c_{\mu,\gamma}^\lambda, c_{\alpha,\nu}^\lambda \neq 0$ and thus the sequences of integer partitions also satisfy equality the following equality constraints

$$|\alpha^{(n)}| + |\beta^{(n)}| = |\mu^{(n)}|, \tag{283}$$

$$|\beta^{(n)}| + |\gamma^{(n)}| = |\nu^{(n)}|, \tag{284}$$

$$|\alpha^{(n)}| + |\beta^{(n)}| + |\gamma^{(n)}| = |\lambda^{(n)}|. \tag{285}$$

C.2 From recoupling to states

In this section, our objective is to prove the converse of the above theorem. Specifically, if the norm of a sequence of recoupling operators with components satisfying Eq. (282) admits of a polynomial lower bound, then there necessarily exists of a configuration of density matrices satisfying Eq. (221). In fact, this converse statement follows from a stronger claim that when there does not exist a configuration of density matrices satisfying the spectral constraints, every tuple of sequences of integer partitions satisfying Eq. (221) has recoupling operator norm eventually decaying at an exponential rate.

Consider the following operator defined by an integral over the normalized Hilbert-Schmidt measure $d\rho$ on density matrices and a given $k \in \mathbb{N}$:

$$\mathbb{E} [\sigma^{\otimes k}] = \int d\sigma \sigma^{\otimes k} \geq \binom{k+d^2-1}{k}^{-1} \rho^{\otimes k} \geq (k+1)^{-(d^2-1)} \rho^{\otimes k}. \tag{286}$$

The three-fold tensor product of the above operator is an integral over $\Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}$ and has the form

$$\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}] := \int d\sigma_1 \int d\sigma_2 \int d\sigma_3 \Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}. \tag{287}$$

The operator $\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}]$ is equivariant with respect to the natural action of

$$S_{k_1} \times S_{k_2} \times S_{k_3} \times U(d) \times U(d) \times U(d). \tag{288}$$

Schur's lemma therefore implies that the operator $\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}]$ is equal to a positive sum of projection operators of the form $\Pi_{\alpha\beta\gamma}$ (where $\alpha \vdash k_1, \beta \vdash k_2$, and $\gamma \vdash k_3$) with the following coefficients:

$$\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}] = \sum_{\alpha, \beta, \gamma} R_{\alpha\beta\gamma} \Pi_{\alpha\beta\gamma} \tag{289}$$

with coefficient $R_{\alpha\beta\gamma}$ given by a ratio of dimensions

$$R_{\alpha\beta\gamma} = \frac{1}{\dim(U_{(k_1)}^{d^2}) \dim(U_{(k_2)}^{d^2}) \dim(U_{(k_3)}^{d^2})} \frac{\dim(U_\alpha^d) \dim(U_\beta^d) \dim(U_\gamma^d)}{\dim(W_\alpha) \dim(W_\beta) \dim(W_\gamma)}. \tag{290}$$

Consequently, $\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}]$ commutes with the projection operators Π_μ, Π_ν and Π_λ and thus also $\Pi_{\alpha\beta\gamma\lambda}^\mu$ and $\Pi_{\alpha\beta\gamma\lambda}^\nu$. In fact the following holds

$$\Pi_{\alpha\beta\gamma\lambda}^\mu \mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}] \Pi_{\alpha\beta\gamma\lambda}^\nu = R_{\alpha\beta\gamma} \Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu \quad (291)$$

Also since operator X acting on a D -dimensional vector space satisfies $\|X\|_\infty \leq \|X\|_{\text{HS}} \leq \sqrt{D} \|X\|_\infty$, one has a polynomial equivalence between the norms of the recoupling symbol

$$\left\| \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right\|_\infty \leq \left\| \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right\|_{\text{HS}} \leq \left(c_{\alpha, \beta}^\mu c_{\beta, \gamma}^\nu c_{\mu, \gamma}^\lambda c_{\alpha, \nu}^\lambda \right)^{\frac{1}{2}} \left\| \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right\|_\infty \quad (292)$$

For the product of projectors, $\Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu$, the only (polynomial or better) bound one has is simply

$$\left\| \Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu \right\|_\infty \leq \left\| \Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu \right\|_{\text{HS}} = \sqrt{\text{Tr}(\Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu)} \quad (293)$$

Using the decomposition provided by Eq. (217), together with Eq. (268) and Eq. (289) yields

$$\left\| \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right\|_{\text{HS}}^2 = \text{Tr} \left(\begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix}^\dagger \begin{Bmatrix} \alpha & \beta & \mu \\ \gamma & \lambda & \nu \end{Bmatrix} \right) \quad (294)$$

$$= \frac{\text{Tr} \left(\Pi_{\alpha\beta\gamma\lambda}^\mu \Pi_{\alpha\beta\gamma\lambda}^\nu \right)}{\dim(U_\lambda^d) \dim(W_\alpha) \dim(W_\beta) \dim(W_\gamma)} \quad (295)$$

$$= Q_{\alpha\beta\gamma\lambda} \text{Tr} \left(\Pi_{\alpha\beta\gamma\lambda}^\mu \mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}] \Pi_{\alpha\beta\gamma\lambda}^\nu \right) \quad (296)$$

$$\leq \sup_{\rho_1 \rho_2 \rho_3} Q_{\alpha\beta\gamma\lambda} \left| \text{Tr} \left(\Pi_{\alpha\beta\gamma\lambda}^\mu \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3} \Pi_{\alpha\beta\gamma\lambda}^\nu \right) \right| \quad (297)$$

$$\leq Q_{\alpha\beta\gamma\lambda} \exp(-k_6 \inf_{\rho_1 \rho_2 \rho_3} I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda} + O(\log(k_6))) \quad (298)$$

where the infimum is taken over all density matrices ρ_1, ρ_2, ρ_3 with $r_i = \text{spec}(\rho_i)$ for $i \in \{1, 2, 3\}$ and r_4, r_5, r_6 the spectra of the appropriate convex combinations of ρ_1, ρ_2, ρ_3 with weights determined by the relative sizes of the integer partitions involved (again here we are making the rather dubious substitution $p_i \approx k_i/k_6$ when substituting the function $I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda}$ **TODO: resolve this issue formally**). Also the scalar $Q_{\alpha\beta\gamma\lambda}$ is

$$Q_{\alpha\beta\gamma\lambda} := \frac{1}{R_{\alpha\beta\gamma} \dim(U_\lambda^d) \dim(W_\alpha) \dim(W_\beta) \dim(W_\gamma)} \quad (299)$$

$$= \frac{\dim(U_{(k_1)}^{d^2}) \dim(U_{(k_2)}^{d^2}) \dim(U_{(k_3)}^{d^2})}{\dim(U_\alpha^d) \dim(U_\beta^d) \dim(U_\gamma^d) \dim(U_\lambda^d)} \quad (300)$$

As we have seen prior, the rate function $I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda}$ only vanishes if all of the spectra match exactly. Furthermore, it can be shown that the infimum over ρ_1, ρ_2, ρ_3 of $I_{r_1, r_2, r_3, r_4, r_5, r_6}^{\alpha, \beta, \gamma, \mu, \nu, \lambda}$ is always bounded below by a constant $I^* > 0$ if there does not exist a configuration of density matrices satisfying the requirements. Consequently, we conclude the following theorem.

Theorem 34. Suppose $(\alpha^{(n)}, \beta^{(n)}, \gamma^{(n)}, \mu^{(n)}, \nu^{(n)}, \lambda^{(n)})$ is a sequence of integer partitions satisfying Eq. (282) for some $0 < p_1, p_2, p_3 < 1$ satisfying $p_1 + p_2 + p_3 = 1$ and $(r_1, r_2, \dots, r_6) \in \Delta_d^{1 \times 6}$. If there does not exist a collection of density matrices $\rho_1, \rho_2, \rho_3, \rho_4, \rho_5$ and ρ_6 with spectra $r_i = \text{spec}(\rho_i)$ satisfying Eq. (221), then there exists a constant $I^* > 0$ such that

$$\left\| \begin{Bmatrix} \alpha^{(n)} & \beta^{(n)} & \mu^{(n)} \\ \gamma^{(n)} & \lambda^{(n)} & \nu^{(n)} \end{Bmatrix} \right\|_\infty \leq \exp(-nI^* + O(\log n)) \quad (301)$$

Remark 13. It appears that I did not make explicit use of the following operator inequalities when proving the above theorem.

$$\mathbb{E} [\sigma^{\otimes k}] \geq (k+1)^{-(d^2-1)} \rho^{\otimes k}. \quad (302)$$

$$\mathbb{E} [\Omega_{\sigma_1, \sigma_2, \sigma_3}^{k_1, k_2, k_3}] \geq ((k_1+1)(k_2+1)(k_3+1))^{-d^2+1} \Omega_{\rho_1, \rho_2, \rho_3}^{k_1, k_2, k_3}. \quad (303)$$

D [Bounding the multinomial distribution?]

Proposition 35. *TODO: It would be desirable to have an explicit lower-bound on the probability for the multinomial distribution event which maximizes the likelihood.*

Proof. The proof uses the version of Stirling's approximation due to Robbins [Rob55] which states

$$\log n! = n \log n + \frac{1}{2} \log n - n + r_n + \frac{1}{2} \log(2\pi) \quad (304)$$

where the remainder term is small for large n and satisfies $(12n+1)^{-1} < r_n < (12n)^{-1}$. Consequently, if $\varepsilon_i := n_i - p_i n$ where $n := \sum_{i=1}^k n_i$, the multinomial probability satisfies

$$\log \Pr(n_1, \dots, n_k | p_1, \dots, p_k) \quad (305)$$

$$= \log \left(\binom{n_1 + \dots + n_k}{n_1, \dots, n_k} p_1^{n_1} \dots p_k^{n_k} \right), \quad (306)$$

$$= \log n! - \sum_{i=1}^k \log n_i! + \sum_{i=1}^k n_i \log p_i \quad (307)$$

$$= \left(n \log n + \frac{1}{2} \log n - n + r_n + \frac{1}{2} \log(2\pi) \right) - \sum_{i=1}^k \left(n_i \log n_i + \frac{1}{2} \log n_i - n_i + r_{n_i} + \frac{1}{2} \log(2\pi) \right) + \sum_{i=1}^k n_i \log p_i \quad (308)$$

$$= \left(n \log n + \frac{1}{2} \log n \right) - \sum_{i=1}^k \left(n_i \log n_i + \frac{1}{2} \log n_i \right) + \sum_{i=1}^k n_i \log p_i + \frac{1-k}{2} \log(2\pi) + (r_n - \sum_{i=1}^k r_{n_i}) \quad (309)$$

$$= \left(n \log n + \frac{1}{2} \log n \right) - \sum_{i=1}^k \left(n_i \log(np_i + \varepsilon_i) + \frac{1}{2} \log(np_i + \varepsilon_i) \right) + \sum_{i=1}^k n_i \log p_i + \frac{1-k}{2} \log(2\pi) + (r_n - \sum_{i=1}^k r_{n_i}) \quad (310)$$

$$(311)$$

□

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